

Qualification Pack



Junior Engineer -Civil (Metro Rail)

QP Code: DMR/LSC/Q0401

Version: 1.0

NSQF Level: 4.5

Delhi Metro Rail Corporation Limited || M6FR+8V5, Depot Entry Road, Metro Vihar, Shastri Park,
Shahdara, Delhi
110053 || email:ckprasad@dmrc.org

Qualification Pack

Contents

DMR/LSC/Q0401: Junior Engineer -Civil (Metro Rail)	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
DMR/LSC/N9901: Implement metro operations training and safety procedures	5
DMR/LSC/N0402: Coordinate the railway track maintenance and inspection	17
DMR/LSC/N0403: Coordinate the maintenance and inspection of LWR and associated track elements	31
DMR/LSC/N0405: Oversee accident and emergency response, track renewal, and implement inspection protocols	43
DMR/LSC/N0401: Use P.Way tools and equipment for track inspection maintenance	52
DMR/LSC/N0406: Coordinate construction, maintenance, and inspection in Metro projects	57
DMR/LSC/N0404: Conduct site visits for the inspection and maintenance of viaducts, stations, and treatment plants	72
DGT/VSQ/N0101: Employability Skills (30 Hours)	78
Assessment Guidelines and Weightage	83
<i>Assessment Guidelines</i>	83
<i>Assessment Weightage</i>	84
Acronyms	85
Glossary	87

Qualification Pack

DMR/LSC/Q0401: Junior Engineer -Civil (Metro Rail)

Brief Job Description

A Junior Engineer - Civil (Metro Rail) is responsible for coordinating railway track maintenance and inspection, including LWR and associated track elements, ensuring adherence to inspection protocols. The individual oversees accident and emergency response, track renewal, and the use of P.Way tools and equipment for track maintenance. The role includes coordinating construction and maintenance activities in Metro projects and conducting site inspections for viaducts, stations, and treatment plants to ensure structural integrity and safety compliance.

Personal Attributes

The individual should have strong leadership, attention to detail, problem-solving skills, and the ability to work under pressure. Good communication and teamwork abilities are essential for coordinating with teams and ensuring compliance with safety standards.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [DMR/LSC/N9901: Implement metro operations training and safety procedures](#)
2. [DMR/LSC/N0402: Coordinate the railway track maintenance and inspection](#)
3. [DMR/LSC/N0403: Coordinate the maintenance and inspection of LWR and associated track elements](#)
4. [DMR/LSC/N0405: Oversee accident and emergency response, track renewal, and implement inspection protocols](#)
5. [DMR/LSC/N0401: Use P.Way tools and equipment for track inspection maintenance](#)
6. [DMR/LSC/N0406: Coordinate construction, maintenance, and inspection in Metro projects](#)
7. [DMR/LSC/N0404: Conduct site visits for the inspection and maintenance of viaducts, stations, and treatment plants](#)
8. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Logistics
--------	-----------

Qualification Pack

Sub-Sector	Transport (Metro Rail)
Occupation	Civil
Country	India
NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	Diploma (3 years Engineering Diploma* / equivalent in Civil or in relevant field)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-4.5-TW-03514-2025-V1-DMRCA
NQR Version	1.0

Qualification Pack

DMR/LSC/N9901: Implement metro operations training and safety procedures

Description

This unit focuses on implementing structured training and safety measures for metro operations, ensuring technical knowledge, system readiness, and emergency preparedness.

Scope

The scope covers the following :

- Training Module Overview
- Safety Overview
- Store Management and Procurement
- Overview of Telecom, Signaling, and Track Systems
- Overview of Traction, SCADA, and E&M Systems
- OCC and Rolling Stock Overview
- Operations and Customer Care
- Practical Visits
- First Aid Training
- Fire Fighting Systems
- Security in Metro Systems
- Metro Rail General Rules 2020
- Safety Circulars and Contingency Plans

Elements and Performance Criteria

Training Module Overview

To be competent, the user/individual on the job must be able to:

- PC1.** follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.
- PC2.** conduct identity verification and maintain attendance records
- PC3.** ensure adherence to training timelines and program deliverables

Safety Overview

To be competent, the user/individual on the job must be able to:

- PC4.** identify the role of the safety department and follow measures for personal, passenger, and system safety
- PC5.** use appropriate Personal Protective Equipment (PPE) in all designated areas
- PC6.** follow the hazard identification and risk assessment processes

Store Management and Procurement

To be competent, the user/individual on the job must be able to:

- PC7.** perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.
- PC8.** monitor inventory levels and ensure timely requisitions for required materials
- PC9.** maintain records of procurement processes and vendor communication

Qualification Pack

Overview of Telecom, Signaling, and Track Systems

To be competent, the user/individual on the job must be able to:

- PC10.** identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks
- PC11.** identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.
- PC12.** identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks
- PC13.** report malfunctions in telecom, signaling, or track systems using the correct reporting protocols

Overview of Traction, SCADA, and E&M Systems

To be competent, the user/individual on the job must be able to:

- PC14.** identify traction power and SCADA components and interpret sectioning diagrams
- PC15.** identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality
- PC16.** report unusual system performance and assist in routine inspections

OCC and Rolling Stock Overview

To be competent, the user/individual on the job must be able to:

- PC17.** familiarize with OCC systems, tools, and the role of OCC controllers
- PC18.** identify rolling stock components and their maintenance procedures
- PC19.** report rolling stock irregularities observed during operations

Operations and Customer Care

To be competent, the user/individual on the job must be able to:

- PC20.** familiarize with station operations, revenue management, and crew control functions
- PC21.** identify public complaint channels, possible causes, and resolve complaints using proper processes
- PC22.** assist in managing passenger inquiries and ensuring customer satisfaction

Practical Visits

To be competent, the user/individual on the job must be able to:

- PC23.** conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations
- PC24.** document key observations during practical visits for knowledge reinforcement
- PC25.** identify operational challenges during visits and suggest improvement areas

First Aid Training

To be competent, the user/individual on the job must be able to:

- PC26.** identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)
- PC27.** follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness
- PC28.** perform artificial respiration, CPR, and safe transportation of injured persons using stretchers
- PC29.** participate in simulated first aid scenarios to enhance preparedness

Fire Fighting Systems

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC30.** identify elements of combustion, classes of fire, and appropriate extinguishing methods
- PC31.** operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants
- PC32.** conduct practical drills and identify fire suppression equipment at metro stations
- PC33.** respond to fire incidents promptly, ensuring passenger and system safety

Security in Metro Systems

To be competent, the user/individual on the job must be able to:

- PC34.** identify security threats, red alert situations, and follow protocols for disaster management
- PC35.** handle theft cases, manage abandoned persons, and lodge FIRs as per protocol
- PC36.** coordinate with security personnel to mitigate security breaches effectively

Metro Rail General Rules 2020

To be competent, the user/individual on the job must be able to:

- PC37.** interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols
- PC38.** follow operational procedures for maintenance, station control duties, and driverless train systems
- PC39.** ensure compliance with rules for safe and efficient train operations

Safety Circulars and Contingency Plans

To be competent, the user/individual on the job must be able to:

- PC40.** implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation
- PC41.** follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises
- PC42.** ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods
- PC43.** manage degraded mode operations, including train movement during interlocking failures
- PC44.** oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections
- PC45.** facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks
- PC46.** execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras
- PC47.** oversee train stabling, depot entry/exit, train movement, and fitness checks
- PC48.** conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detainment of passengers due to technical failure, explosion, or fire
- PC49.** operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train
- PC50.** coordinate vehicle movement on link lines and rescue immobilized trains
- PC51.** respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety
- PC52.** operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of punctuality, discipline, and structured training
- KU2.** the procedures for trainee verification and attendance record maintenance
- KU3.** the role of time management in achieving training milestones
- KU4.** the role of the safety department and station/depot safety protocols
- KU5.** the use and importance of PPE
- KU6.** the basics of hazard identification, risk management, and accident reporting
- KU7.** the basics of stores management (inventory control, documentation)
- KU8.** the procurement processes and coordination across departments
- KU9.** the procedures for vendor management and maintaining procurement transparency
- KU10.** the telecom components, including AFC, CCTV, clocks, and communication systems
- KU11.** the basics of signaling systems (CATC, CBTC) and reporting unusual signal observations
- KU12.** the key P-Way components, track patrolling, and reporting track conditions
- KU13.** the common malfunctions in telecom, signaling, and track systems
- KU14.** the traction power systems and sectioning diagrams
- KU15.** the importance of SCADA and its functionalities
- KU16.** the role of E&M systems (DG sets, UPS) and their use during emergencies
- KU17.** the key preventive maintenance checks for E&M systems
- KU18.** the OCC systems, tools, and controller responsibilities
- KU19.** the rolling stock components, features, and basic maintenance
- KU20.** the reporting and addressing of common rolling stock issues
- KU21.** the station operations, revenue management, and crew coordination
- KU22.** the causes of public complaints and grievance resolution processes
- KU23.** the techniques for effective customer service and passenger engagement
- KU24.** the observations and learning outcomes from OCC, station, depot, and tunnel visits
- KU25.** the identifying real-world challenges during visits and linking theory to practice
- KU26.** the basic principles of first aid and key human body functions
- KU27.** the management of bleeding, fractures, burns, and unconsciousness
- KU28.** the artificial respiration, CPR, and safe patient transportation
- KU29.** the importance of first aid preparedness during emergencies
- KU30.** the elements of combustion, fire classifications, and extinguishing methods
- KU31.** the principles of passive/active fire protection systems
- KU32.** the practical usage of fire extinguishers, hydrants, and drills
- KU33.** the emergency response procedures during fire incidents
- KU34.** the security protocols, red alert situations, and theft management processes
- KU35.** the protocols for managing abandoned persons and disaster situations
- KU36.** the role of security personnel in ensuring passenger and asset safety
- KU37.** the key terminologies, signal usage, and operational protocols

Qualification Pack

- KU38.** the rules for maintenance works, train operations, and driverless train systems
- KU39.** the role of regulatory compliance in metro rail safety
- KU40.** the safety circulars and contingency plans for fire/security incidents
- KU41.** the importance of proactive contingency planning for emergency scenarios.
- KU42.** the importance role-playing and drills to improve safety and response performance
- KU43.** the procedures for TSR imposition, axle counter reset, Train ID management, and CATS operations
- KU44.** the engineering possession protocols for maintenance tasks during non-revenue hours, including solar system testing and hot work
- KU45.** the guidelines for safely halting train operations during emergencies, natural disasters, or technical failures
- KU46.** the steps for managing degraded mode operations and interlocking failures
- KU47.** the SOP and SWO/DWO lifecycle management, including revisions and timetable trials for new sections
- KU48.** the ATO, UTO, and ATB mode operations, emphasizing operational readiness and AFTC checks
- KU49.** the procedures for neutral section negotiation, pantograph damage management, and monitoring Panto-OHE interfaces using cameras
- KU50.** the best practices for train stabling, depot operations, and train fitness verification
- KU51.** the emergency evacuation protocols, focusing on passenger safety in UG and mid-tunnel situations
- KU52.** the guidelines for operating non-ATP vehicles, handling rail fractures, weld failures, and resetting alarms during rollbacks
- KU53.** the vehicle operations on link lines and immobilized train rescue procedures
- KU54.** the protocols for fire/smoke incidents, particularly UG fires, and ensuring safety during such incidents
- KU55.** the operation of ETS/BLS plungers and removal of foreign objects from OHE systems, along with OHE safety practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** convey clear and concise information to teams, junior engineers, and customers
- GS2.** effectively prioritize tasks and adhere to training schedules and operational timelines
- GS3.** identify operational challenges and propose practical solutions during training or real-time operations
- GS4.** assess system performance, identify malfunctions, and report anomalies accurately
- GS5.** work cohesively with peers, instructors, and junior engineers to meet training objectives
- GS6.** address passenger queries and manage grievances professionally
- GS7.** proactively identify risks, follow safety protocols, and contribute to hazard-free operations
- GS8.** adapt to diverse training environments, system layouts, and evolving operational scenarios
- GS9.** accurately maintain logs, records, and reports for training and practical activities

Logo
Not
Available



Qualification Pack

- GS10.** monitor systems, equipment, and processes, identifying areas requiring intervention or improvement

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Training Module Overview</i>	2	2	-	1
PC1. follow schedules for weekly activities and guidelines on punctuality, discipline, and conduct.	-	-	-	-
PC2. conduct identity verification and maintain attendance records	-	-	-	-
PC3. ensure adherence to training timelines and program deliverables	-	-	-	-
<i>Safety Overview</i>	2	2	-	1
PC4. identify the role of the safety department and follow measures for personal, passenger, and system safety	-	-	-	-
PC5. use appropriate Personal Protective Equipment (PPE) in all designated areas	-	-	-	-
PC6. follow the hazard identification and risk assessment processes	-	-	-	-
<i>Store Management and Procurement</i>	2	2	-	1
PC7. perform store management activities, including stock handling, documentation, and interdepartmental coordination for procurement.	-	-	-	-
PC8. monitor inventory levels and ensure timely requisitions for required materials	-	-	-	-
PC9. maintain records of procurement processes and vendor communication	-	-	-	-
<i>Overview of Telecom, Signaling, and Track Systems</i>	2	2	-	1
PC10. identify telecom components (AFC, CCTV, PIDS, radio systems, clocks) and perform basic functional checks	-	-	-	-
PC11. identify signaling systems (CATC, CBTC) and observe/report signal functioning during visits.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify P-Way components (rails, sleepers, crossings) and report track conditions through basic patrolling tasks	-	-	-	-
PC13. report malfunctions in telecom, signaling, or track systems using the correct reporting protocols	-	-	-	-
<i>Overview of Traction, SCADA, and E&M Systems</i>	2	2	-	1
PC14. identify traction power and SCADA components and interpret sectioning diagrams	-	-	-	-
PC15. identify E&M systems (RSS, ASS, DG sets, UPS) and perform basic checks to understand their functionality	-	-	-	-
PC16. report unusual system performance and assist in routine inspections	-	-	-	-
<i>OCC and Rolling Stock Overview</i>	2	2	-	1
PC17. familiarize with OCC systems, tools, and the role of OCC controllers	-	-	-	-
PC18. identify rolling stock components and their maintenance procedures	-	-	-	-
PC19. report rolling stock irregularities observed during operations	-	-	-	-
<i>Operations and Customer Care</i>	2	2	-	1
PC20. familiarize with station operations, revenue management, and crew control functions	-	-	-	-
PC21. identify public complaint channels, possible causes, and resolve complaints using proper processes	-	-	-	-
PC22. assist in managing passenger inquiries and ensuring customer satisfaction	-	-	-	-
<i>Practical Visits</i>	1	1	-	0.5
PC23. conduct visits to OCC, stations, depots, and tunnels to observe layouts, facilities, and operations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. document key observations during practical visits for knowledge reinforcement	-	-	-	-
PC25. identify operational challenges during visits and suggest improvement areas	-	-	-	-
<i>First Aid Training</i>	1	1	-	0.5
PC26. identify the role and scope of first aid, including key human body functions (e.g., circulatory, respiratory systems)	-	-	-	-
PC27. follow appropriate techniques for treating bleeding, fractures, burns, shock, and unconsciousness	-	-	-	-
PC28. perform artificial respiration, CPR, and safe transportation of injured persons using stretchers	-	-	-	-
PC29. participate in simulated first aid scenarios to enhance preparedness	-	-	-	-
<i>Fire Fighting Systems</i>	1	1	-	0.5
PC30. identify elements of combustion, classes of fire, and appropriate extinguishing methods	-	-	-	-
PC31. operate active/passive fire protection systems, including alarms, fire extinguishers, and fire hydrants	-	-	-	-
PC32. conduct practical drills and identify fire suppression equipment at metro stations	-	-	-	-
PC33. respond to fire incidents promptly, ensuring passenger and system safety	-	-	-	-
<i>Security in Metro Systems</i>	1	1	-	0.5
PC34. identify security threats, red alert situations, and follow protocols for disaster management	-	-	-	-
PC35. handle theft cases, manage abandoned persons, and lodge FIRs as per protocol	-	-	-	-
PC36. coordinate with security personnel to mitigate security breaches effectively	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Metro Rail General Rules 2020</i>	1	1	-	0.5
PC37. interpret key provisions under Metro Rail General Rules 2020, including signals, train speed limits, and safety protocols	-	-	-	-
PC38. follow operational procedures for maintenance, station control duties, and driverless train systems	-	-	-	-
PC39. ensure compliance with rules for safe and efficient train operations	-	-	-	-
<i>Safety Circulars and Contingency Plans</i>	1	1	-	0.5
PC40. implement procedures for the imposition of TSR, axle counter reset, Train ID management, and Computer Aided Train Supervision (CATS) operation	-	-	-	-
PC41. follow guidelines for engineering possession and maintenance activities during non-revenue hours, including hot work, solar system testing/repair, and role-play exercises	-	-	-	-
PC42. ensure compliance with procedures for stopping trains at platforms and during operational disruptions, such as earthquakes, high winds, and floods	-	-	-	-
PC43. manage degraded mode operations, including train movement during interlocking failures	-	-	-	-
PC44. oversee the issue, revision, updating, and deletion of SOPs, SWO/DWO, and timetable trials for new sections	-	-	-	-
PC45. facilitate the initiation of revenue train services, including pilot train movement, ATO/UTO/ATB mode operations, and AFTC functionality checks	-	-	-	-
PC46. execute procedures for negotiating neutral sections, managing pantograph failures, and monitoring Panto-OHE interface cameras	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC47. oversee train stabling, depot entry/exit, train movement, and fitness checks	-	-	-	-
PC48. conduct emergency evacuation using UG facilities, mid-tunnel shafts, and walkways for detrainment of passengers due to technical failure, explosion, or fire	-	-	-	-
PC49. operate and manage non-ATP vehicles such as RGM and CMV/CTMC and address incidents like rail fractures, weld failures, rollback scenarios, radio communication issues and resetting of passenger alarms in train	-	-	-	-
PC50. coordinate vehicle movement on link lines and rescue immobilized trains	-	-	-	-
PC51. respond to fire/smoke incidents, including UG train fires, ensuring passenger and staff safety	-	-	-	-
PC52. operate ETS/BLS plungers, remove foreign objects like kites from OHE, and follow OHE safety guidelines	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N9901
NOS Name	Implement metro operations training and safety procedures
Sector	Logistics
Sub-Sector	
Occupation	Foundation – Metro (Rail)
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DMR/LSC/N0402: Coordinate the railway track maintenance and inspection

Description

This unit outlines the skills and knowledge required to perform inspection, maintenance, and repairs on railway track components, including rails, joints, ballast, sleepers, turnouts, and both ballasted and ballastless track systems. It focuses on ensuring track safety, stability, and operational efficiency through proper handling, assessment, and documentation of track elements and their conditions.

Scope

The scope covers the following :

- Rails and Joints
- Ballastless Track
- Formation & Ballast
- Sleepers & Fastenings
- Turnouts -Assembly, Layouts & Inspection
- Ballastless Track Maintenance
- Ballasted Track Maintenance

Elements and Performance Criteria

Rails and Joints

To be competent, the user/individual on the job must be able to:

- PC1.** identify and document the type and condition of rails (90 UTS and HH) during inspections, ensuring they meet required standards for grade and service life
- PC2.** follow and implement safe and correct techniques for loading, unloading, and stacking high UTS and HH rails to prevent damage and ensure safety
- PC3.** measure rail wear accurately and report any wear exceeding permissible limits to schedule timely maintenance or replacement
- PC4.** carry out lubrication of outer rails on curves either manually or by operating electro-mechanical devices, ensuring even and adequate coverage
- PC5.** check the electronic lubrication system's performance, including lubricant levels and sensor function, and report any discrepancies for corrective action
- PC6.** inspect rails for corrosion and undertake anti-corrosion treatment where needed, documenting the treatment process for record-keeping
- PC7.** inspect and secure fish-plates, fish-bolts, and joggled fish-plates, ensuring proper alignment and tightness to maintain joint integrity
- PC8.** monitor and record rail temperature changes and assess any required adjustments to maintain safe expansion gaps according to environmental conditions
- PC9.** inspect expansion joints for damage or misalignment, ensuring they allow for proper rail expansion and contraction, and report any issues needing maintenance
- PC10.** examine insulated glued joints for signs of wear or separation and confirm that alignment meets standards for track stability and signaling

Qualification Pack

Ballast less Track

To be competent, the user/individual on the job must be able to:

- PC11.** demonstrate an understanding of the principles of ballast less track design, including key features, benefits, and challenges compared to traditional ballasted tracks
- PC12.** implement the key steps for constructing ballastless tracks using both top-down and bottom-up methods
- PC13.** oversee and document routine maintenance activities on ballastless track, such as inspecting track alignment, fastening systems, and expansion joints, to ensure track integrity and safety
- PC14.** inspect for the correct installation and alignment of different elastic fastening systems (such as Fastening System 336 and Fastening System 300) to ensure proper load distribution
- PC15.** measure and document the toe load of fastening systems (including System 336, System 300, Fastening, and Rheda 2000) to conform to ballastless track application standards
- PC16.** inspect the plinth beam profile and anti-derailment arrangements in tunnels and viaducts, ensuring they comply with design specifications for safe rail operations

Formation & Ballast

To be competent, the user/individual on the job must be able to:

- PC17.** inspect track formation to ensure it meets specified design and structural requirements, including stability and load-bearing capacity
- PC18.** check the functions of track formation, focusing on load distribution, stability, and support for track components
- PC19.** check the typical cross-sectional elements of formation on embankments for single and double lines on Broad Gauge (BG), noting any deviations from the standard profile
- PC20.** inspect and maintain the drainage system around the track formation, ensuring it effectively diverts water to prevent erosion or structural damage
- PC21.** measure and verify the slope on top of the formation, ensuring it meets design specifications for optimal drainage and stability
- PC22.** inspect side and catch water drains for blockages or damage, ensuring they are clear and functional to prevent water accumulation on the formation
- PC23.** identify common causes of formation failure, such as poor drainage or soil erosion, and report findings for preventive maintenance planning
- PC24.** coordinate remedial actions on weak formations, including cess repair, side slope correction, and cess widening, following standard procedures for stabilization
- PC25.** inspect ballast sections on BG tracks, including straights and curves, to verify they meet specifications for single and double lines on Long Welded Rail (LWR) sections
- PC26.** collect and test track ballast samples as per specifications, documenting measurements to confirm ballast quality and adherence to standards

Sleepers & Fastenings

To be competent, the user/individual on the job must be able to:

- PC27.** inspect concrete sleepers for any signs of damage or wear, ensuring they meet structural requirements for load-bearing and track stability
- PC28.** check the installation and alignment of mono-block pre-stressed sleepers, ensuring they are correctly positioned and securely fastened

Qualification Pack

- PC29.** apply the understanding of the general properties and advantages of different types of sleepers, focusing on durability, stability, and compatibility with various track conditions
- PC30.** inspect special sleepers installed at level crossings for proper alignment and secure fastening, ensuring smooth and safe vehicular passage
- PC31.** check turnout sleepers for correct alignment and fastening to ensure they accommodate track switching without compromising stability
- PC32.** examine special sleepers at Switch Expansion Joints (SEJ) locations, bridge approaches, and sharp curves to verify they meet design requirements for flexibility and support
- PC33.** measure and verify rail coning on sleepers, ensuring it meets standards to maintain rail stability and minimize lateral movement
- PC34.** implement proper handling and stacking techniques for Pre-stressed Concrete (PSC) sleepers to prevent damage during transportation and storage
- PC35.** inspect inserts in PSC sleepers for wear or damage, ensuring they provide secure anchorage for rail fastenings
- PC36.** install and inspect Elastic Rail Clips (MK-III) on PSC sleepers to confirm they provide sufficient rail restraint and elasticity
- PC37.** measure the toe load of rail clips and document findings to verify they meet the specified requirements for rail restraint
- PC38.** assess the condition of fastenings against renewal criteria and recommend replacement when fastenings show signs of fatigue or reduced effectiveness
- PC39.** inspect and oversee the replacement of rubber pads under rails on PSC sleepers, ensuring they provide adequate cushioning and noise reduction
- PC40.** examine rail liners for wear or displacement, and replace them if necessary to maintain track alignment and reduce rail seat wear

Turnouts - Assembly, Layouts & Inspection

To be competent, the user/individual on the job must be able to:

- PC41.** assemble and align turnouts with 1 in 8½, 1 in 12, 1 in 9, and 1 in 7 configurations, ensuring all components are correctly positioned according to specified dimensions and standards
- PC42.** measure and verify the main dimensions of assembled turnouts, including rail gauge, spacing, and angle, to ensure adherence to design specifications
- PC43.** identify and differentiate contrary and similar flexure turnouts (1 in 8½, 1 in 12, 1 in 9, and 1 in 7) based on layout and function, ensuring they are assembled correctly
- PC44.** explain and identify components and configurations of diamond crossings and scissors crossovers, accurately using terminology in reference to track layout
- PC45.** inspect and maintain switch components for proper alignment, movement, and secure fastening to facilitate smooth switching operations
- PC46.** inspect diamond crossings for wear, alignment, and stability, ensuring they meet operational standards for intersecting tracks
- PC47.** lay and maintain fan-shaped turnouts by aligning components precisely and conducting routine maintenance to prevent wear and misalignment
- PC48.** perform maintenance on turnouts with Cast Manganese Steel (CMS) crossings, inspecting for wear or cracking and ensuring secure fastening to the track base
- PC49.** conduct detailed inspections of various turnout types, recording measurements, identifying defects, and ensuring components meet operational tolerances

Qualification Pack

PC50. complete inspection formats accurately for each type of turnout, documenting track condition, alignment, and any deviations from standard parameters

PC51. measure and confirm that turnout track parameters, such as gauge, alignment, and elevation, fall within the specified tolerance limits

Ballast less Track Maintenance

To be competent, the user/individual on the job must be able to:

PC52. conduct regular patrolling of ballastless track to identify any immediate issues or safety hazards, such as debris, cracks, or misalignments, and report findings promptly

PC53. measure key track parameters (gauge, cross-level, twist, versines, alignment, wear) during inspections, recording data to ensure compliance with track tolerance standards

PC54. measure and document toe loads on rail clips and fastenings to verify they meet specified requirements for securing rails on ballastless track

PC55. inspect points and crossings for wear, alignment, and secure fastening, identifying any defects that require maintenance or adjustment

PC56. verify that track measurements (gauge, alignment, cross-level) fall within approved tolerances using standard inspection methods, and document results accurately

PC57. inspect the reinforced concrete plinth (anti-derailment beam) for cracks, spalling, or misalignment, and perform repairs as necessary to maintain structural integrity

PC58. adjust gauge, alignment, and cross-level using eccentric insulating bushes and shim plates to ensure the track meets specified dimensions and tolerances

PC59. oversee the replacement of worn or damaged ballastless elastic fastenings and elastomeric pads to maintain track stability and minimize vibrations

PC60. assess track condition based on wear, age, and environmental factors to determine if renewal is required, following specified guidelines for ballastless track maintenance

Ballasted Track Maintenance

To be competent, the user/individual on the job must be able to:

PC61. inspect the gauge face of rails for wear and, where necessary, carry out gauge-face reversal to extend rail life, ensuring secure reassembly and alignment

PC62. replace CMS crossings as needed, following safe handling procedures, securing fastenings, and ensuring proper alignment and track stability

PC63. replace worn or damaged rails by removing the existing rail, positioning the new rail accurately, and fastening it securely to maintain gauge and alignment

PC64. replace base plates on ballast tracks by lifting the rail, aligning and securing the new base plate, and ensuring proper seating and fastening

PC65. replace the second drive mechanism in point machines, ensuring proper alignment and secure connections to maintain precise point and crossing operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the historical background of advent of metros in India, and their role as a Mass Transit System in the Urban Transport System

KU2. the advantages and disadvantages of metro systems and their economics vis-a-vis other conventional systems of surface transportation

Qualification Pack

- KU3.** the maintenance wing of Metro
- KU4.** different types of tracks, their purpose, and maintenance requirements
- KU5.** the concept of metro lines, their classification (e.g., mainline, branch line), and operational protocols
- KU6.** the role of depots in maintenance, storage, and cleaning of metro trains
- KU7.** standard sections, grade & service life of 90 UTS and HH rails, along with proper methods for careful loading, unloading, stacking, and handling of rails
- KU8.** the causes and permissible limits of rail wear and precautions for high UTS/HH rails
- KU9.** the lubrication of outer rail of curves manually and using electro-mechanical or electronic rail lubrication systems
- KU10.** corrosion of rails, anti-corrosion treatments, and expansion/contraction of rails with temperature, including joints like fish-plates, insulated glued joints, and expansion joints for bridges
- KU11.** the concepts and principles of constructing ballastless tracks using top-down and bottom-up methods
- KU12.** the maintenance activities, including elastic fastenings (e.g., Fastening System 336, Fastening System 300), plinth beam profiles, and anti-derailment arrangements in tunnels and viaducts
- KU13.** typical sections of formation on embankments, causes of formation failure, and remedial actions such as cess repair and side slope corrections
- KU14.** ballast sections on BG tracks for straights and curves, specifications, testing, and measurement of track ballast
- KU15.** general aspects of concrete sleepers, including mono-block pre-stressed sleepers, special sleepers for level crossings, turnouts, SEJs, and bridge approaches
- KU16.** the handling precautions and fastenings for PSC sleepers, including inserts, elastic rail clip MK-III, toe load measurement, rubber pads, liners, and criteria for fastening renewal
- KU17.** turnouts, including components, types, wear limits of CMS crossings, and curved/thick web/derailing switches
- KU18.** assembly and maintenance of turnouts (e.g. 1 in 8 1/2, 1 in 12) and diamond crossings, including fan-shaped turnouts and detailed inspection formats
- KU19.** the regular inspections and patrolling of ballastless tracks, measuring track parameters (e.g., gauge, cross-level, twist, versines, alignments, wear), and approved methods of inspection
- KU20.** the inspection and repair of reinforced concrete plinths, correction of gauge and alignment using eccentric bushes, shim plates, and renewal of ballastless elastic fastenings and elastomeric pads
- KU21.** the factors governing track renewals, gauge-face reversal, and replacement of heavy track components like CMS crossings, rails, base plates, and second drives

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate inspection findings, maintenance requirements, and operational updates clearly and concisely to team members, junior engineers, and stakeholders

Qualification Pack

- GS2.** collaborate with cross-functional teams to ensure the seamless execution of tasks such as track maintenance, alignment verification, and safety inspections
- GS3.** identify issues like rail wear, misalignment, or system malfunctions and devise appropriate corrective actions to maintain track safety and integrity
- GS4.** assess track conditions, prioritize maintenance activities, and make timely decisions to minimize disruptions and enhance operational efficiency
- GS5.** plan and execute tasks such as routine inspections, repairs, and documentation within set timelines to ensure uninterrupted rail operations
- GS6.** measure and record track parameters, inspecting critical components, and documenting maintenance activities to meet operational standards
- GS7.** adjust to changing environmental conditions or emergencies during track inspections and maintenance activities, ensuring safety and compliance
- GS8.** implement workplace safety protocols and practices to minimize risks during maintenance operations, particularly when handling heavy equipment or working in hazardous conditions
- GS9.** work effectively within a team, ensuring tasks such as joint inspections, component replacements, and system checks are performed collaboratively and efficiently

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Rails and Joints</i>	4	4	-	2
PC1. identify and document the type and condition of rails (90 UTS and HH) during inspections, ensuring they meet required standards for grade and service life	-	-	-	-
PC2. follow and implement safe and correct techniques for loading, unloading, and stacking high UTS and HH rails to prevent damage and ensure safety	-	-	-	-
PC3. measure rail wear accurately and report any wear exceeding permissible limits to schedule timely maintenance or replacement	-	-	-	-
PC4. carry out lubrication of outer rails on curves either manually or by operating electro-mechanical devices, ensuring even and adequate coverage	-	-	-	-
PC5. check the electronic lubrication system's performance, including lubricant levels and sensor function, and report any discrepancies for corrective action	-	-	-	-
PC6. inspect rails for corrosion and undertake anti-corrosion treatment where needed, documenting the treatment process for record-keeping	-	-	-	-
PC7. inspect and secure fish-plates, fish-bolts, and joggled fish-plates, ensuring proper alignment and tightness to maintain joint integrity	-	-	-	-
PC8. monitor and record rail temperature changes and assess any required adjustments to maintain safe expansion gaps according to environmental conditions	-	-	-	-
PC9. inspect expansion joints for damage or misalignment, ensuring they allow for proper rail expansion and contraction, and report any issues needing maintenance	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. examine insulated glued joints for signs of wear or separation and confirm that alignment meets standards for track stability and signaling	-	-	-	-
<i>Ballast less Track</i>	2	2	-	1
PC11. demonstrate an understanding of the principles of ballast less track design, including key features, benefits, and challenges compared to traditional ballasted tracks	-	-	-	-
PC12. implement the key steps for constructing ballastless tracks using both top-down and bottom-up methods	-	-	-	-
PC13. oversee and document routine maintenance activities on ballastless track, such as inspecting track alignment, fastening systems, and expansion joints, to ensure track integrity and safety	-	-	-	-
PC14. inspect for the correct installation and alignment of different elastic fastening systems (such as Fastening System 336 and Fastening System 300) to ensure proper load distribution	-	-	-	-
PC15. measure and document the toe load of fastening systems (including System 336, System 300, Fastening, and Rheda 2000) to conform to ballastless track application standards	-	-	-	-
PC16. inspect the plinth beam profile and anti-derailment arrangements in tunnels and viaducts, ensuring they comply with design specifications for safe rail operations	-	-	-	-
<i>Formation & Ballast</i>	3	3	-	1
PC17. inspect track formation to ensure it meets specified design and structural requirements, including stability and load-bearing capacity	-	-	-	-
PC18. check the functions of track formation, focusing on load distribution, stability, and support for track components	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. check the typical cross-sectional elements of formation on embankments for single and double lines on Broad Gauge (BG), noting any deviations from the standard profile	-	-	-	-
PC20. inspect and maintain the drainage system around the track formation, ensuring it effectively diverts water to prevent erosion or structural damage	-	-	-	-
PC21. measure and verify the slope on top of the formation, ensuring it meets design specifications for optimal drainage and stability	-	-	-	-
PC22. inspect side and catch water drains for blockages or damage, ensuring they are clear and functional to prevent water accumulation on the formation	-	-	-	-
PC23. identify common causes of formation failure, such as poor drainage or soil erosion, and report findings for preventive maintenance planning	-	-	-	-
PC24. coordinate remedial actions on weak formations, including cess repair, side slope correction, and cess widening, following standard procedures for stabilization	-	-	-	-
PC25. inspect ballast sections on BG tracks, including straights and curves, to verify they meet specifications for single and double lines on Long Welded Rail (LWR) sections	-	-	-	-
PC26. collect and test track ballast samples as per specifications, documenting measurements to confirm ballast quality and adherence to standards	-	-	-	-
<i>Sleepers & Fastenings</i>	4	4	-	2
PC27. inspect concrete sleepers for any signs of damage or wear, ensuring they meet structural requirements for load-bearing and track stability	-	-	-	-
PC28. check the installation and alignment of mono-block pre-stressed sleepers, ensuring they are correctly positioned and securely fastened	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. apply the understanding of the general properties and advantages of different types of sleepers, focusing on durability, stability, and compatibility with various track conditions	-	-	-	-
PC30. inspect special sleepers installed at level crossings for proper alignment and secure fastening, ensuring smooth and safe vehicular passage	-	-	-	-
PC31. check turnout sleepers for correct alignment and fastening to ensure they accommodate track switching without compromising stability	-	-	-	-
PC32. examine special sleepers at Switch Expansion Joints (SEJ) locations, bridge approaches, and sharp curves to verify they meet design requirements for flexibility and support	-	-	-	-
PC33. measure and verify rail coning on sleepers, ensuring it meets standards to maintain rail stability and minimize lateral movement	-	-	-	-
PC34. implement proper handling and stacking techniques for Pre-stressed Concrete (PSC) sleepers to prevent damage during transportation and storage	-	-	-	-
PC35. inspect inserts in PSC sleepers for wear or damage, ensuring they provide secure anchorage for rail fastenings	-	-	-	-
PC36. install and inspect Elastic Rail Clips (MK-III) on PSC sleepers to confirm they provide sufficient rail restraint and elasticity	-	-	-	-
PC37. measure the toe load of rail clips and document findings to verify they meet the specified requirements for rail restraint	-	-	-	-
PC38. assess the condition of fastenings against renewal criteria and recommend replacement when fastenings show signs of fatigue or reduced effectiveness	-	-	-	-
PC39. inspect and oversee the replacement of rubber pads under rails on PSC sleepers, ensuring they provide adequate cushioning and noise reduction	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC40. examine rail liners for wear or displacement, and replace them if necessary to maintain track alignment and reduce rail seat wear	-	-	-	-
<i>Turnouts - Assembly, Layouts & Inspection</i>	3	3	-	2
PC41. assemble and align turnouts with 1 in 8½, 1 in 12, 1 in 9, and 1 in 7 configurations, ensuring all components are correctly positioned according to specified dimensions and standards	-	-	-	-
PC42. measure and verify the main dimensions of assembled turnouts, including rail gauge, spacing, and angle, to ensure adherence to design specifications	-	-	-	-
PC43. identify and differentiate contrary and similar flexure turnouts (1 in 8½, 1 in 12, 1 in 9, and 1 in 7) based on layout and function, ensuring they are assembled correctly	-	-	-	-
PC44. explain and identify components and configurations of diamond crossings and scissors crossovers, accurately using terminology in reference to track layout	-	-	-	-
PC45. inspect and maintain switch components for proper alignment, movement, and secure fastening to facilitate smooth switching operations	-	-	-	-
PC46. inspect diamond crossings for wear, alignment, and stability, ensuring they meet operational standards for intersecting tracks	-	-	-	-
PC47. lay and maintain fan-shaped turnouts by aligning components precisely and conducting routine maintenance to prevent wear and misalignment	-	-	-	-
PC48. perform maintenance on turnouts with Cast Manganese Steel (CMS) crossings, inspecting for wear or cracking and ensuring secure fastening to the track base	-	-	-	-
PC49. conduct detailed inspections of various turnout types, recording measurements, identifying defects, and ensuring components meet operational tolerances	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC50. complete inspection formats accurately for each type of turnout, documenting track condition, alignment, and any deviations from standard parameters	-	-	-	-
PC51. measure and confirm that turnout track parameters, such as gauge, alignment, and elevation, fall within the specified tolerance limits	-	-	-	-
<i>Ballast less Track Maintenance</i>	3	3	-	1
PC52. conduct regular patrolling of ballastless track to identify any immediate issues or safety hazards, such as debris, cracks, or misalignments, and report findings promptly	-	-	-	-
PC53. measure key track parameters (gauge, cross-level, twist, versines, alignment, wear) during inspections, recording data to ensure compliance with track tolerance standards	-	-	-	-
PC54. measure and document toe loads on rail clips and fastenings to verify they meet specified requirements for securing rails on ballastless track	-	-	-	-
PC55. inspect points and crossings for wear, alignment, and secure fastening, identifying any defects that require maintenance or adjustment	-	-	-	-
PC56. verify that track measurements (gauge, alignment, cross-level) fall within approved tolerances using standard inspection methods, and document results accurately	-	-	-	-
PC57. inspect the reinforced concrete plinth (anti-derailment beam) for cracks, spalling, or misalignment, and perform repairs as necessary to maintain structural integrity	-	-	-	-
PC58. adjust gauge, alignment, and cross-level using eccentric insulating bushes and shim plates to ensure the track meets specified dimensions and tolerances	-	-	-	-
PC59. oversee the replacement of worn or damaged ballastless elastic fastenings and elastomeric pads to maintain track stability and minimize vibrations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC60. assess track condition based on wear, age, and environmental factors to determine if renewal is required, following specified guidelines for ballastless track maintenance	-	-	-	-
<i>Ballasted Track Maintenance</i>	1	1	-	1
PC61. inspect the gauge face of rails for wear and, where necessary, carry out gauge-face reversal to extend rail life, ensuring secure reassembly and alignment	-	-	-	-
PC62. replace CMS crossings as needed, following safe handling procedures, securing fastenings, and ensuring proper alignment and track stability	-	-	-	-
PC63. replace worn or damaged rails by removing the existing rail, positioning the new rail accurately, and fastening it securely to maintain gauge and alignment	-	-	-	-
PC64. replace base plates on ballast tracks by lifting the rail, aligning and securing the new base plate, and ensuring proper seating and fastening	-	-	-	-
PC65. replace the second drive mechanism in point machines, ensuring proper alignment and secure connections to maintain precise point and crossing operations	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0402
NOS Name	Coordinate the railway track maintenance and inspection
Sector	Logistics
Sub-Sector	
Occupation	Civil
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DMR/LSC/N0403: Coordinate the maintenance and inspection of LWR and associated track elements

Description

This unit focuses on maintaining safety, reliability, and efficient management of electrical systems in traction power operations. Key competencies include educating personnel on safety protocols, conducting risk assessments, operating equipment, troubleshooting issues, and ensuring compliance with regulatory standards through coordinated efforts with internal teams and external authorities.

Scope

The scope covers the following :

- Long Welded Rails (LWR)
- Welding of Rails
- Rail Flaws & Testing
- Horizontal Curves
- Vertical Curves
- Reconditioning of CMS Crossings & Switches
- Mechanized Maintenance
- Use of Trolley/Lorry/Dolly
- Protection, Restrictions & Indications

Elements and Performance Criteria

Long Welded Rails (LWR)

To be competent, the user/individual on the job must be able to:

- PC1.** conduct regular patrolling and inspections, focusing on measuring track parameters (gauge, cross-level, twist, versines, alignment, wear), and recording findings to ensure the track is within standard tolerances
- PC2.** measure toe loads on fastenings and inspect points and crossings for wear and secure alignment, ensuring all components function safely and meet required specifications
- PC3.** verify that track tolerances, including gauge and alignment, fall within approved limits during inspections, noting any deviations and arranging for corrective actions as necessary
- PC4.** perform seasonal maintenance by conducting pre-monsoon inspections, monitoring during the monsoon for water accumulation, and conducting post-monsoon checks to address any erosion or drainage issues on the LWR track
- PC5.** adhere to limits for maximum lifting/lowering per session when lifting or lowering the track, install ramps at track ends, and verify adequate ballast cushioning and drainage post-adjustment
- PC6.** perform conventional through packing or use off-track machines to compact ballast effectively around sleepers, ensuring stability and alignment of the LWR track
- PC7.** conduct overhauling and deep screening of ballast to remove fine materials and restore ballast support, ensuring improved drainage and track stability

Qualification Pack

- PC8.** inspect and maintain special track layouts, including points and crossings, ensuring components such as crossings and switch rails are aligned, secured, and function effectively
- PC9.** perform scheduled periodic inspections and programmed maintenance attentions as per the LWR maintenance guidelines, documenting track conditions and scheduling necessary repairs

Welding of Rails

To be competent, the user/individual on the job must be able to:

- PC10.** use different types of rail thermometers, identify their role in monitoring rail temperature for safe LWR/CWR (Continuous Welded Rail/Long Welded Rail) maintenance
- PC11.** verify and select permitted locations for laying LWR/CWR, ensuring compliance with specified guidelines on track structure and environmental conditions
- PC12.** inspect the track structure of LWR/CWR, focusing on securing components such as fastenings and ballast, and ensuring alignment for stability under stress conditions
- PC13.** perform de-stressing operations with or without rail tensors, ensuring that rails are stress-free at the desired temperature to prevent buckling or fracture
- PC14.** identify and repair rail fractures and cases of buckling by applying standard procedures to restore track integrity and safety
- PC15.** follow maintenance precautions for CWR/LWR, including correct fastening, inspection, and environmental adjustments, to prevent rail stress-related issues
- PC16.** conduct patrolling in both hot and cold weather, monitoring rail conditions and expansion/contraction, and reporting any anomalies for immediate action
- PC17.** determine and confirm the stress-free temperature for rail sections using established guidelines, ensuring that LWR/CWR remains stable across temperature variations
- PC18.** inspect LWR sections, especially gaps at Switch Expansion Joints (SEJ), and implement corrective measures to ensure they meet the required gap tolerances

Rail Flaws & Testing

To be competent, the user/individual on the job must be able to:

- PC19.** identify common types of rail and weld defects, such as cracks, corrosion, and misalignment, and report these defects accurately for further inspection and maintenance
- PC20.** perform a visual examination of rails and welds during patrols, identifying surface-level defects and anomalies, and documenting findings for scheduled maintenance
- PC21.** conduct Ultrasonic Flaw Detection (USFD) testing on rails and welds to detect internal flaws, interpreting results to determine the need for repair or replacement

Horizontal Curves

To be competent, the user/individual on the job must be able to:

- PC22.** identify and classify the degree and types of curves on a section of track, verifying compliance with design specifications and logging details accurately in the curve register
- PC23.** measure and calculate the relationship between degree, radius, chord, and versine on curves, ensuring they align with specified geometric standards for safe train operations
- PC24.** calculate and verify the super elevation (equilibrium cant), maximum cant, cant deficiency, and cant excess for curves, adjusting as necessary to optimize train stability and comfort
- PC25.** apply the formula for safe and equilibrium speed on curves, determining permissible speeds based on existing cant, radius, and track conditions

Qualification Pack

- PC26.** calculate the required cant for a given curve and the corresponding permissible speed, ensuring adjustments align with track safety standards
- PC27.** determine the transition length and maximum cant gradient, confirming that transitions are smooth and meet specifications for track geometry on curves
- PC28.** measure and adjust the rate of running out of cant and cant deficiency, verifying that the rate aligns with specified standards to maintain smooth curve transitions
- PC29.** inspect and implement required widening of gauge and extra clearance on curves, adjusting to accommodate lateral forces and prevent excessive wear on rail components
- PC30.** verify that the minimum required length of straight track between reverse curves is provided, ensuring smooth transitions and maintaining train stability

Vertical Curves

To be competent, the user/individual on the job must be able to:

- PC31.** identify the need for vertical curves at gradient changes to provide a smooth transition for trains, minimizing stress on rails and ensuring passenger comfort
- PC32.** calculate the equivalent radius for vertical curves to assess curvature impact on train movement, ensuring it meets specified safety and comfort standards
- PC33.** inspect and maintain vertical curves specifically on LWR sections, ensuring that the curves are stable, within tolerance limits, and do not induce excessive stress on welded rails

Reconditioning of CMS Crossings & Switches

To be competent, the user/individual on the job must be able to:

- PC34.** assess and recondition worn-out Cast Manganese Steel (CMS) crossings and switches by identifying worn areas, preparing surfaces, and restoring profiles to extend service life and maintain safe track conditions
- PC35.** select and use the appropriate type of electrodes for welding and reconditioning CMS crossings and switches, ensuring compatibility with CMS material to achieve durable and safe repairs

Mechanized Maintenance

To be competent, the user/individual on the job must be able to:

- PC36.** use various small track machines for maintenance tasks, to improve efficiency and safety in track upkeep
- PC37.** apply off-track tampers to provide spot attention on sections requiring immediate alignment correction, ensuring quick and accurate adjustments to maintain track stability
- PC38.** verify all pre-requisites for machine tamping, including securing clearance, checking track geometry, and ensuring equipment functionality before tamping work begins
- PC39.** perform pre-tamping inspections, monitor during tamping operations, and conduct post-tamping checks with the Unimat-Compact to confirm track stability and alignment
- PC40.** identify Rail Contact Fatigue (RCF) defects and corrugation on rail surfaces, understanding the need for regular inspections to prevent severe rail degradation
- PC41.** measure and record RCF defects and corrugation on rails using appropriate inspection tools, ensuring accurate data collection for maintenance planning
- PC42.** operate, inspect, and maintain the Rail Grinding Machine (RGM) to address RCF defects and corrugation, ensuring effective grinding for extended rail life and smooth track surface

Use of Trolley/Lorry/Dolly

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC43.** operate push trolleys, lorries, and motor trolleys safely, considering their specific functions, capacities, and limitations for on-track operations
- PC44.** ensure all required safety equipment is available and functional before deploying any trolley or lorry, implementing protection procedures to prevent track obstructions
- PC45.** operate rail dollies for the movement of rails and materials, following protection protocols to secure the track section during dolly use and prevent interference with train operations

Protection, Restrictions & Indications

To be competent, the user/individual on the job must be able to:

- PC46.** use hand signals with flags during the day and lamps at night to communicate effectively with train operators and other track workers, ensuring clear and consistent signaling
- PC47.** deploy banner flags and engineering indicators accurately to mark work areas or hazards, ensuring appropriate placement of temporary and permanent indicators
- PC48.** implement appropriate protection measures during emergencies, such as setting up caution orders and ensuring warning signals and indicators are visible to approaching trains
- PC49.** apply appropriate speed restrictions for various track attention activities/failures, ensuring trains are informed of and adhere to safe operational speeds

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the patrolling, regular inspections, and measurement of track parameters including gauge, cross-level, twist, versines, alignments, wear, and toe loads
- KU2.** point and crossing inspection, track tolerances for ballasted track, and pre-monsoon, monsoon, and post-monsoon attentions
- KU3.** the guidelines for lifting and lowering tracks, including maximum permissible lifting/lowering at one time, provision of ramps, adequate ballast cushion, and drainage after lowering
- KU4.** conventional through packing, off-track machines, overhauling, deep screening, and maintenance of special layouts such as points and crossings
- KU5.** the definitions, types of rail thermometers, and locations for laying LWR/CWR
- KU6.** de-stressing with and without rail tensors, maintenance precautions for LWR/CWR, cold & hot weather patrolling, and stress-free temperature determination
- KU7.** inspection of LWR and remedial actions for correcting gaps at SEJ, as well as repairs for rail fractures and buckling
- KU8.** the types of rail and weld defects, visual examination by patrolmen, and ultrasonic flaw detection (USFD) testing of rails and welds
- KU9.** the degree of curve, relation between degree, radius, chord, and versine, and recording in P.Way's curve register
- KU10.** the concepts of super elevation, equilibrium cant, cant deficiency, cant excess, permissible speed on curves, and calculation of transition length
- KU11.** the widening of gauge on curves, extra clearance, minimum length of straight between reverse curves, vertical curves on LWR, and grades with equivalent radius
- KU12.** the reconditioning of worn-out crossings and switches, including types of electrodes used
- KU13.** the importance of mechanization, small track machines, their types, and uses in maintenance, and pre/post requirements for machine tamping by Unimat-Compact

Qualification Pack

- KU14.** Rail contact fatigue (RCF) defects, corrugation, identification, and measurement of defects
- KU15.** the working of Rail Grinding Machine (RGM), inspection, and maintenance of RGM defects
- KU16.** the working of push trolley, lorry, motor trolley, and rail dolly, including equipment, protection, and operational guidelines
- KU17.** the use of hand signals, hand flags, hand lamps, banner flags, and engineering indicators for day and night operations
- KU18.** temporary and permanent protection measures for emergencies, issuance of caution orders, and speed restrictions to be followed during track attention or failures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record and document measurements, inspections, and maintenance activities accurately
- GS2.** communicate effectively using hand signals, flags, lamps, and verbal instructions
- GS3.** perform calculations and analyze track parameters to ensure safety and compliance
- GS4.** identify and resolve issues related to track defects, alignment, or rail stress
- GS5.** operate specialized track maintenance tools and equipment proficiently
- GS6.** plan and prioritize maintenance tasks to balance routine and emergency activities
- GS7.** pay attention to detail when verifying tolerances, inspecting flaws, and monitoring conditions
- GS8.** follow safety protocols and use PPE to ensure a safe working environment
- GS9.** collaborate with team members and junior engineers for efficient track maintenance
- GS10.** adapt to new technologies and methods to improve track maintenance practices

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Long Welded Rails (LWR)</i>	3	3	-	2
PC1. conduct regular patrolling and inspections, focusing on measuring track parameters (gauge, cross-level, twist, versines, alignment, wear), and recording findings to ensure the track is within standard tolerances	-	-	-	-
PC2. measure toe loads on fastenings and inspect points and crossings for wear and secure alignment, ensuring all components function safely and meet required specifications	-	-	-	-
PC3. verify that track tolerances, including gauge and alignment, fall within approved limits during inspections, noting any deviations and arranging for corrective actions as necessary	-	-	-	-
PC4. perform seasonal maintenance by conducting pre-monsoon inspections, monitoring during the monsoon for water accumulation, and conducting post-monsoon checks to address any erosion or drainage issues on the LWR track	-	-	-	-
PC5. adhere to limits for maximum lifting/lowering per session when lifting or lowering the track, install ramps at track ends, and verify adequate ballast cushioning and drainage post-adjustment	-	-	-	-
PC6. perform conventional through packing or use off-track machines to compact ballast effectively around sleepers, ensuring stability and alignment of the LWR track	-	-	-	-
PC7. conduct overhauling and deep screening of ballast to remove fine materials and restore ballast support, ensuring improved drainage and track stability	-	-	-	-
PC8. inspect and maintain special track layouts, including points and crossings, ensuring components such as crossings and switch rails are aligned, secured, and function effectively	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. perform scheduled periodic inspections and programmed maintenance attentions as per the LWR maintenance guidelines, documenting track conditions and scheduling necessary repairs	-	-	-	-
<i>Welding of Rails</i>	3	3	-	1
PC10. use different types of rail thermometers, identify their role in monitoring rail temperature for safe LWR/CWR (Continuous Welded Rail/Long Welded Rail) maintenance	-	-	-	-
PC11. verify and select permitted locations for laying LWR/CWR, ensuring compliance with specified guidelines on track structure and environmental conditions	-	-	-	-
PC12. inspect the track structure of LWR/CWR, focusing on securing components such as fastenings and ballast, and ensuring alignment for stability under stress conditions	-	-	-	-
PC13. perform de-stressing operations with or without rail tensors, ensuring that rails are stress-free at the desired temperature to prevent buckling or fracture	-	-	-	-
PC14. identify and repair rail fractures and cases of buckling by applying standard procedures to restore track integrity and safety	-	-	-	-
PC15. follow maintenance precautions for CWR/LWR, including correct fastening, inspection, and environmental adjustments, to prevent rail stress-related issues	-	-	-	-
PC16. conduct patrolling in both hot and cold weather, monitoring rail conditions and expansion/contraction, and reporting any anomalies for immediate action	-	-	-	-
PC17. determine and confirm the stress-free temperature for rail sections using established guidelines, ensuring that LWR/CWR remains stable across temperature variations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC18. inspect LWR sections, especially gaps at Switch Expansion Joints (SEJ), and implement corrective measures to ensure they meet the required gap tolerances	-	-	-	-
<i>Rail Flaws & Testing</i>	2	2	-	1
PC19. identify common types of rail and weld defects, such as cracks, corrosion, and misalignment, and report these defects accurately for further inspection and maintenance	-	-	-	-
PC20. perform a visual examination of rails and welds during patrols, identifying surface-level defects and anomalies, and documenting findings for scheduled maintenance	-	-	-	-
PC21. conduct Ultrasonic Flaw Detection (USFD) testing on rails and welds to detect internal flaws, interpreting results to determine the need for repair or replacement	-	-	-	-
<i>Horizontal Curves</i>	3	3	-	1
PC22. identify and classify the degree and types of curves on a section of track, verifying compliance with design specifications and logging details accurately in the curve register	-	-	-	-
PC23. measure and calculate the relationship between degree, radius, chord, and versine on curves, ensuring they align with specified geometric standards for safe train operations	-	-	-	-
PC24. calculate and verify the super elevation (equilibrium can't), maximum cant, can't deficiency, and can't excess for curves, adjusting as necessary to optimize train stability and comfort	-	-	-	-
PC25. apply the formula for safe and equilibrium speed on curves, determining permissible speeds based on existing cant, radius, and track conditions	-	-	-	-
PC26. calculate the required cant for a given curve and the corresponding permissible speed, ensuring adjustments align with track safety standards	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. determine the transition length and maximum cant gradient, confirming that transitions are smooth and meet specifications for track geometry on curves	-	-	-	-
PC28. measure and adjust the rate of running out of cant and cant deficiency, verifying that the rate aligns with specified standards to maintain smooth curve transitions	-	-	-	-
PC29. inspect and implement required widening of gauge and extra clearance on curves, adjusting to accommodate lateral forces and prevent excessive wear on rail components	-	-	-	-
PC30. verify that the minimum required length of straight track between reverse curves is provided, ensuring smooth transitions and maintaining train stability	-	-	-	-
<i>Vertical Curves</i>	2	2	-	1
PC31. identify the need for vertical curves at gradient changes to provide a smooth transition for trains, minimizing stress on rails and ensuring passenger comfort	-	-	-	-
PC32. calculate the equivalent radius for vertical curves to assess curvature impact on train movement, ensuring it meets specified safety and comfort standards	-	-	-	-
PC33. inspect and maintain vertical curves specifically on LWR sections, ensuring that the curves are stable, within tolerance limits, and do not induce excessive stress on welded rails	-	-	-	-
<i>Reconditioning of CMS Crossings & Switches</i>	2	2	-	1
PC34. assess and recondition worn-out Cast Manganese Steel (CMS) crossings and switches by identifying worn areas, preparing surfaces, and restoring profiles to extend service life and maintain safe track conditions	-	-	-	-
PC35. select and use the appropriate type of electrodes for welding and reconditioning CMS crossings and switches, ensuring compatibility with CMS material to achieve durable and safe repairs	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Mechanized Maintenance</i>	2	2	-	1
PC36. use various small track machines for maintenance tasks, to improve efficiency and safety in track upkeep	-	-	-	-
PC37. apply off-track tampers to provide spot attention on sections requiring immediate alignment correction, ensuring quick and accurate adjustments to maintain track stability	-	-	-	-
PC38. verify all pre-requisites for machine tamping, including securing clearance, checking track geometry, and ensuring equipment functionality before tamping work begins	-	-	-	-
PC39. perform pre-tamping inspections, monitor during tamping operations, and conduct post-tamping checks with the Unimat-Compact to confirm track stability and alignment	-	-	-	-
PC40. identify Rail Contact Fatigue (RCF) defects and corrugation on rail surfaces, understanding the need for regular inspections to prevent severe rail degradation	-	-	-	-
PC41. measure and record RCF defects and corrugation on rails using appropriate inspection tools, ensuring accurate data collection for maintenance planning	-	-	-	-
PC42. operate, inspect, and maintain the Rail Grinding Machine (RGM) to address RCF defects and corrugation, ensuring effective grinding for extended rail life and smooth track surface	-	-	-	-
<i>Use of Trolley/Lorry/Dolly</i>	2	2	-	1
PC43. operate push trolleys, lorries, and motor trolleys safely, considering their specific functions, capacities, and limitations for on-track operations	-	-	-	-
PC44. ensure all required safety equipment is available and functional before deploying any trolley or lorry, implementing protection procedures to prevent track obstructions	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC45. operate rail dollies for the movement of rails and materials, following protection protocols to secure the track section during dolly use and prevent interference with train operations	-	-	-	-
<i>Protection, Restrictions & Indications</i>	1	1	-	1
PC46. use hand signals with flags during the day and lamps at night to communicate effectively with train operators and other track workers, ensuring clear and consistent signaling	-	-	-	-
PC47. deploy banner flags and engineering indicators accurately to mark work areas or hazards, ensuring appropriate placement of temporary and permanent indicators	-	-	-	-
PC48. implement appropriate protection measures during emergencies, such as setting up caution orders and ensuring warning signals and indicators are visible to approaching trains	-	-	-	-
PC49. apply appropriate speed restrictions for various track attention activities/failures, ensuring trains are informed of and adhere to safe operational speeds	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0403
NOS Name	Coordinate the maintenance and inspection of LWR and associated track elements
Sector	Logistics
Sub-Sector	
Occupation	Civil
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DMR/LSC/N0405: Oversee accident and emergency response, track renewal, and implement inspection protocols

Description

This unit specifies the competencies required to handle accidents, unusual occurrences, and emergencies on the railway track. It covers the protocols for track renewals, systematic inspections, and compliance with the Schedule of Dimensions (SoD). It also includes requirements for record-keeping, material handling, and the use of advanced tools for monitoring track conditions. The unit emphasizes safety, efficiency, and adherence to standards in all related activities.

Scope

The scope covers the following :

- Accidents & Unusual Occurrences
- Track Renewals
- Inspections, Track Recording & Monitoring
- Schedule of Dimensions (SoD)
- Inspection Schedule of Engineers and Junior Engineers

Elements and Performance Criteria

Accidents & Unusual Occurrences

To be competent, the user/individual on the job must be able to:

- PC1.** follow Metro's Accident Rules (2002) for accident reporting, ensuring accurate documentation and immediate communication of accidents to relevant authorities
- PC2.** coordinate the duties of P-Way officials during an accident, including securing the site, assessing track damage, setting up protection measures, etc.
- PC3.** implement appropriate safety protocols like securing the area and communicating with control centers during unusual occurrences e.g. fire, smoke, flooding, or track obstructions
- PC4.** assess and mark the area of defective tracks, and arrange for necessary repairs
- PC5.** ensure safe and timely track restoration after accidents or buckling incidents
- PC6.** maintain records of accidents and unusual occurrences, documenting all relevant details, actions taken, and lessons learned for future reference and compliance
- PC7.** inspect and ensure that emergency materials and equipment are readily available and in good condition for immediate use in track restoration or accident response situations
- PC8.** analyze past accident and incident cases in the P-Way wing, identifying lessons learned and best practices to improve future response strategies

Track Renewals

To be competent, the user/individual on the job must be able to:

- PC9.** execute casual and complete track renewal procedures by identifying sections requiring renewal, preparing necessary materials, and ensuring work completion to maintenance standards

Qualification Pack

- PC10.** perform through rail renewals by assessing track segments against un-serviceability criteria, determining sections that require replacement, and conducting renewal activities to restore track integrity
- PC11.** replace worn or damaged elastic pads and fastenings as part of track renewal, ensuring secure and proper installation to maintain track stability
- PC12.** ensure proper handling and storage of track materials (rails, pads, fastenings) during the renewal process to prevent damage and maintain them in optimal condition
- PC13.** apply appropriate speed restrictions during track renewal activities, coordinating with train operations to ensure safe passage of trains near work zones

Inspections, Track Recording & Monitoring

To be competent, the user/individual on the job must be able to:

- PC14.** conduct inspections with the objective of assessing track safety and riding quality, identifying any irregularities that may affect passenger comfort or train stability
- PC15.** measure and evaluate the riding index using established metrics to ensure the track meets standards for smooth train operation and passenger comfort
- PC16.** operate an oscillograph car and interpret Oscillation Monitoring System (OMS) data to assess track geometry and detect irregularities that require attention
- PC17.** use Calipri and Miniprof tools to measure rail wear and profile, ensuring accurate data collection to inform maintenance decisions
- PC18.** conduct track geometry inspections with a track geometry trolley, capturing data on alignment, gauge, and level to maintain track quality standards

Schedule of Dimensions (SoD)

To be competent, the user/individual on the job must be able to:

- PC19.** identify and apply the key dimensions and clearance requirements specific to Broad Gauge (1676 mm) in the Metro SoD to ensure compliance during maintenance and construction activities
- PC20.** identify and apply the Standard Gauge (1435 mm) SoD specifications, ensuring track and infrastructure elements adhere to these dimensions for safe and efficient operations on standard gauge sections
- PC21.** implement the specific SoD requirements for Standard Gauge (1435 mm) on the Airport Line, considering additional factors such as platform alignment and structural clearances tailored for this line
- PC22.** apply the SoD specifications for Standard Gauge (1435 mm) with Wide Rolling Stock, ensuring that infrastructure and track components provide sufficient clearance for wider vehicles on the designated lines

Inspection Schedule of Engineers and Junior Engineers

To be competent, the user/individual on the job must be able to:

- PC23.** conduct regular foot and cab inspections of the track to assess the condition of rails, sleepers, and fastenings, identifying any immediate or potential issues for prompt maintenance action
- PC24.** inspect curves for alignment, gauge, and wear to ensure they meet safety and operational standards, taking corrective actions if deviations are observed
- PC25.** inspect points and crossings for proper alignment, wear, and functionality, ensuring safe movement across junctions and implementing adjustments or repairs as needed

Qualification Pack

- PC26.** perform detailed inspections of Long Welded Rail (LWR) and Continuous Welded Rail (CWR) track sections, checking for track stability and any signs of rail stress or buckling
- PC27.** accompany track recording and OMS operations to observe data collection, assist with inspections, and address any identified irregularities in real-time
- PC28.** oversee the execution of works affecting track conditions, ensuring that all maintenance and construction work complies with safety standards and has minimal operational disruption
- PC29.** implement emergency response actions by securing the track, coordinating with relevant teams, and ensuring that appropriate protection and caution measures are in place
- PC30.** maintain and organize stores and inventory, ensuring that all essential track materials and equipment are available, in good condition, and correctly documented
- PC31.** update all relevant inspection and maintenance registers regularly, ensuring accurate reporting and compliance with Metro standards for documentation and record-keeping

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Metro's Accidents Rules-2002, reporting of accidents, and duties of P.Way officials during accidents
- KU2.** the actions to be taken during unusual occurrences such as fire and smoke, flooding, and obstructions on the track
- KU3.** the procedures for defective track reporting, restoration of track after accidents, and restoration in cases of buckling
- KU4.** the records of accidents/unusual occurrences and materials required for emergencies
- KU5.** case studies on past experiences and lessons learned in the P.Way wing to improve response and preventive measures
- KU6.** the processes for casual renewals, complete track renewals, through rail renewals, and criteria for un-serviceability
- KU7.** the renewal of elastic pads and fastenings, careful handling of materials, and speed restrictions during renewal activities
- KU8.** the objectives of inspection including riding quality and riding index
- KU9.** the methods of inspection using tools like Oscillograph Car, OMS, Calipri, and Miniprof applications
- KU10.** inspection by Track Geometry Trolley for detailed analysis
- KU11.** the salient features of different types of Schedule of Dimensions (SOD) including, Broad Gauge (1676 mm), Standard Gauge (1435 mm), Standard Gauge (1435 mm) for Airport Line, Standard Gauge (1435 mm) with wide rolling stock
- KU12.** the procedures for foot inspection, cab inspection, and checking of curves
- KU13.** the inspection of points and crossings, LWR/CWR track, and accompanying track recordings using OMS
- KU14.** the execution of works affecting track, actions in case of emergencies, and upkeeping of store and inventory
- KU15.** the requirement of updating relevant registers of sections, reporting procedures, and ensuring compliance

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** document and report accidents per metro's Accident Rules (2002)
- GS2.** oversee site security, track damage assessment, and protection measures during accidents
- GS3.** implement safety measures for fire, flooding, and track obstructions
- GS4.** arrange repairs for defective tracks and ensure timely restoration
- GS5.** maintain emergency equipment for track restoration and accident response
- GS6.** coordinate train operations with speed limits near work zones
- GS7.** conduct inspections to identify irregularities affecting safety or comfort
- GS8.** measure and evaluate the riding quality to ensure operational standards
- GS9.** use tools like OMS and track geometry trolleys to assess alignment and gauge
- GS10.** maintain accurate records of inspections, maintenance, and unusual occurrences

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Accidents & Unusual Occurrences</i>	5	5	-	2
PC1. follow Metro's Accident Rules (2002) for accident reporting, ensuring accurate documentation and immediate communication of accidents to relevant authorities	-	-	-	-
PC2. coordinate the duties of P-Way officials during an accident, including securing the site, assessing track damage, setting up protection measures, etc.	-	-	-	-
PC3. implement appropriate safety protocols like securing the area and communicating with control centers during unusual occurrences e.g. fire, smoke, flooding, or track obstructions	-	-	-	-
PC4. assess and mark the area of defective tracks, and arrange for necessary repairs	-	-	-	-
PC5. ensure safe and timely track restoration after accidents or buckling incidents	-	-	-	-
PC6. maintain records of accidents and unusual occurrences, documenting all relevant details, actions taken, and lessons learned for future reference and compliance	-	-	-	-
PC7. inspect and ensure that emergency materials and equipment are readily available and in good condition for immediate use in track restoration or accident response situations	-	-	-	-
PC8. analyze past accident and incident cases in the P-Way wing, identifying lessons learned and best practices to improve future response strategies	-	-	-	-
<i>Track Renewals</i>	4	4	-	2
PC9. execute casual and complete track renewal procedures by identifying sections requiring renewal, preparing necessary materials, and ensuring work completion to maintenance standards	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. perform through rail renewals by assessing track segments against un-serviceability criteria, determining sections that require replacement, and conducting renewal activities to restore track integrity	-	-	-	-
PC11. replace worn or damaged elastic pads and fastenings as part of track renewal, ensuring secure and proper installation to maintain track stability	-	-	-	-
PC12. ensure proper handling and storage of track materials (rails, pads, fastenings) during the renewal process to prevent damage and maintain them in optimal condition	-	-	-	-
PC13. apply appropriate speed restrictions during track renewal activities, coordinating with train operations to ensure safe passage of trains near work zones	-	-	-	-
<i>Inspections, Track Recording & Monitoring</i>	4	4	-	2
PC14. conduct inspections with the objective of assessing track safety and riding quality, identifying any irregularities that may affect passenger comfort or train stability	-	-	-	-
PC15. measure and evaluate the riding index using established metrics to ensure the track meets standards for smooth train operation and passenger comfort	-	-	-	-
PC16. operate an oscillograph car and interpret Oscillation Monitoring System (OMS) data to assess track geometry and detect irregularities that require attention	-	-	-	-
PC17. use Calipri and Miniprof tools to measure rail wear and profile, ensuring accurate data collection to inform maintenance decisions	-	-	-	-
PC18. conduct track geometry inspections with a track geometry trolley, capturing data on alignment, gauge, and level to maintain track quality standards	-	-	-	-
<i>Schedule of Dimensions (SoD)</i>	3	3	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. identify and apply the key dimensions and clearance requirements specific to Broad Gauge (1676 mm) in the Metro SoD to ensure compliance during maintenance and construction activities	-	-	-	-
PC20. identify and apply the Standard Gauge (1435 mm) SoD specifications, ensuring track and infrastructure elements adhere to these dimensions for safe and efficient operations on standard gauge sections	-	-	-	-
PC21. implement the specific SoD requirements for Standard Gauge (1435 mm) on the Airport Line, considering additional factors such as platform alignment and structural clearances tailored for this line	-	-	-	-
PC22. apply the SoD specifications for Standard Gauge (1435 mm) with Wide Rolling Stock, ensuring that infrastructure and track components provide sufficient clearance for wider vehicles on the designated lines	-	-	-	-
<i>Inspection Schedule of Engineers and Junior Engineers</i>	4	4	-	2
PC23. conduct regular foot and cab inspections of the track to assess the condition of rails, sleepers, and fastenings, identifying any immediate or potential issues for prompt maintenance action	-	-	-	-
PC24. inspect curves for alignment, gauge, and wear to ensure they meet safety and operational standards, taking corrective actions if deviations are observed	-	-	-	-
PC25. inspect points and crossings for proper alignment, wear, and functionality, ensuring safe movement across junctions and implementing adjustments or repairs as needed	-	-	-	-
PC26. perform detailed inspections of Long Welded Rail (LWR) and Continuous Welded Rail (CWR) track sections, checking for track stability and any signs of rail stress or buckling	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. accompany track recording and OMS operations to observe data collection, assist with inspections, and address any identified irregularities in real-time	-	-	-	-
PC28. oversee the execution of works affecting track conditions, ensuring that all maintenance and construction work complies with safety standards and has minimal operational disruption	-	-	-	-
PC29. implement emergency response actions by securing the track, coordinating with relevant teams, and ensuring that appropriate protection and caution measures are in place	-	-	-	-
PC30. maintain and organize stores and inventory, ensuring that all essential track materials and equipment are available, in good condition, and correctly documented	-	-	-	-
PC31. update all relevant inspection and maintenance registers regularly, ensuring accurate reporting and compliance with Metro standards for documentation and record-keeping	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0405
NOS Name	Oversee accident and emergency response, track renewal, and implement inspection protocols
Sector	Logistics
Sub-Sector	
Occupation	Civil
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DMR/LSC/N0401: Use P.Way tools and equipment for track inspection maintenance

Description

This unit focuses on the identification, proper use, and maintenance of tools and equipment essential for P.Way tasks, including inspections, maintenance, and specialized operations like AT welding, sleeper packing, and track geometry assessment. It emphasizes safety, precision, and compliance with railway standards in handling tools for point and crossing, curves, SEJs, buffer stops, and P.Way materials.

Scope

The scope covers the following :

- P.Way Tools and Equipment

Elements and Performance Criteria

P-Way Tools and Equipment

To be competent, the user/individual on the job must be able to:

- PC1.** identify and correctly use tools and equipment specific to point and crossing inspection, e.g. gauge measuring tools, track jacks, and alignment bars, ensuring accurate measurement and alignment checks during maintenance operations
- PC2.** identify and properly use tools and equipment for curve and SEJ inspection, e.g. versine measurement tools, alignment gauges, and thermometers, to accurately assess curve alignment, super-elevation, and SEJ functionality
- PC3.** identify and accurately use tools and equipment for measuring toe load and ensure the correct installation and maintenance of ballastless and ballasted track fittings
- PC4.** identify and correctly use tools and equipment required for buffer stop inspection, including measurement tapes for distance checks, alignment bars for positional accuracy, and torque wrenches for securing fastenings
- PC5.** operate the tools and equipment specific to Oscillation Monitoring System (OMS), Ultrasonic Flaw Detection (USFD), and Miniprof applications, using them to measure track geometry, detect internal rail flaws, and assess rail profiles for wear and alignment
- PC6.** identify and use the Calipri measurement tool and the track geometry trolley, to record track parameters like rail profile, gauge, cross-level, and alignment, ensuring that track geometry meets specified tolerances and safety standards
- PC7.** identify and correctly use tools and equipment, including lifting slings, clamps, and the gantry crane, to safely handle, lift, and position P.Way materials such as rails, sleepers, and fastenings
- PC8.** operate tools and equipment for greasing curves, including manual and automated greasing machines, applying appropriate lubrication to the outer rail of curves to reduce friction and wear, while ensuring even and controlled application
- PC9.** use appropriate tools and equipment for packing sleepers, such as ballast tamping tools, track jacks, and packing hammers, to compact ballast under sleepers, ensuring proper support, alignment, and stability of the track

Qualification Pack

- PC10.** identify and correctly operate tools and equipment specific to AT welding, including crucibles, molds, pre-heating torches, and grinding machines, to safely and effectively execute rail welding, ensuring strong, defect-free welds

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the tools and equipment for point and crossing inspection, curve inspection, SEJ inspection, and their operation during inspection activities
- KU2.** how to use the appropriate tools and equipment for buffer stop inspection
- KU3.** the tools and equipment for toe load measurement, ballastless, and ballasted fittings, and their operational procedures
- KU4.** the tools like OMS, USFD, and Miniprof application, and their operation during inspection
- KU5.** tools for Calipri measurement and inspection using the track geometry trolley
- KU6.** the tools and equipment for handling P.Way materials, including the use of gantries during operation
- KU7.** the tools and equipment for greasing of curves and operation of greasing machines
- KU8.** tools and techniques for packing sleepers effectively
- KU9.** tools and equipment for AT welding and their correct usage during operation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** troubleshoot and resolve issues during track inspections and maintenance activities
- GS2.** convey information effectively and coordinate with team members and other departments
- GS3.** ensure accuracy in measurements, inspections, and maintenance documentation
- GS4.** work effectively in a team environment to ensure seamless track maintenance operations
- GS5.** prioritize tasks and meet deadlines for inspections, repairs, and reporting
- GS6.** adjust to new tools, technologies, or procedures as per operational requirements
- GS7.** assess unusual situations and implement suitable responses to ensure safety and efficiency
- GS8.** document findings, actions taken, and recommendations in clear and precise reports
- GS9.** adhere to safety protocols and ensure compliance with operational guidelines
- GS10.** learn new concepts, tools, and methodologies to enhance job performance

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>P-Way Tools and Equipment</i>	20	20	-	10
PC1. identify and correctly use tools and equipment specific to point and crossing inspection, e.g. gauge measuring tools, track jacks, and alignment bars, ensuring accurate measurement and alignment checks during maintenance operations	-	-	-	-
PC2. identify and properly use tools and equipment for curve and SEJ inspection, e.g. versine measurement tools, alignment gauges, and thermometers, to accurately assess curve alignment, super-elevation, and SEJ functionality	-	-	-	-
PC3. identify and accurately use tools and equipment for measuring toe load and ensure the correct installation and maintenance of ballastless and ballasted track fittings	-	-	-	-
PC4. identify and correctly use tools and equipment required for buffer stop inspection, including measurement tapes for distance checks, alignment bars for positional accuracy, and torque wrenches for securing fastenings	-	-	-	-
PC5. operate the tools and equipment specific to Oscillation Monitoring System (OMS), Ultrasonic Flaw Detection (USFD), and Miniprof applications, using them to measure track geometry, detect internal rail flaws, and assess rail profiles for wear and alignment	-	-	-	-
PC6. identify and use the Calipri measurement tool and the track geometry trolley, to record track parameters like rail profile, gauge, cross-level, and alignment, ensuring that track geometry meets specified tolerances and safety standards	-	-	-	-
PC7. identify and correctly use tools and equipment, including lifting slings, clamps, and the gantry crane, to safely handle, lift, and position P.Way materials such as rails, sleepers, and fastenings	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. operate tools and equipment for greasing curves, including manual and automated greasing machines, applying appropriate lubrication to the outer rail of curves to reduce friction and wear, while ensuring even and controlled application	-	-	-	-
PC9. use appropriate tools and equipment for packing sleepers, such as ballast tamping tools, track jacks, and packing hammers, to compact ballast under sleepers, ensuring proper support, alignment, and stability of the track	-	-	-	-
PC10. identify and correctly operate tools and equipment specific to AT welding, including crucibles, molds, pre-heating torches, and grinding machines, to safely and effectively execute rail welding, ensuring strong, defect-free welds	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0401
NOS Name	Use P.Way tools and equipment for track inspection maintenance
Sector	Logistics
Sub-Sector	
Occupation	Civil, Civil
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DMR/LSC/N0406: Coordinate construction, maintenance, and inspection in Metro projects

Description

This unit provides a comprehensive understanding of construction methodologies, maintenance practices, and inspection standards specific to Metro operations, including underground and elevated structures. It covers viaducts, bridges, tunnels, station buildings, water supply systems, sanitation, and horticulture, ensuring adherence to safety, quality, and operational excellence.

Scope

The scope covers the following :

- Overview of Construction Activities in Metro
- Metro Station Building - Elevated & Underground
- Duties and Responsibilities of Works Junior Engineer
- Viaducts/Bridge/Foundation/Steel Structure - Inspection & Maintenance
- Horticulture Works in O&M
- Maintenance of Stations Elevated and Underground
- Maintenance of Tunnels
- Concrete Technology and Pre-stressed Concrete
- Metro Building Specifications
- Water Supply
- Sanitation
- PTM Pipeline
- Estimates
- Tender and Contract for Works and Standard Specifications

Elements and Performance Criteria

Overview of Construction Activities in Metro

To be competent, the user/individual on the job must be able to:

- PC1.** identify the main construction stages and components of underground and elevated metro stations, including excavation, foundation laying, and superstructure construction
- PC2.** identify the appropriate tools and equipment for constructing viaducts and bridges, including U girders, single and twin segmental box girders, and I-girders
- PC3.** identify special spans, including cantilever and extra dosed bridge spans, and the specific equipment and support structures required to construct and stabilize the spans
- PC4.** apply the cut-and-cover method and top-bottom and bottom-up techniques, understanding their applications in constructing metro tunnels within densely populated urban environments
- PC5.** identify the basic operations of Tunnel Boring Machines (TBMs), including the equipment setup, tunneling stages, and safety protocols essential for efficient and safe TBM usage in metro tunneling

Qualification Pack

- PC6.** apply the New Austrian Tunneling Method (NATM) in controlled ground excavation and use shotcrete for tunnel support, ensuring stability and safety in tunnel construction

Metro Station Building - Elevated & Underground

To be competent, the user/individual on the job must be able to:

- PC7.** identify the functions and access points of key components of elevated and underground Metro stations, including system rooms like the Station Master Room (SMR), Central Control Center (CCC), Emergency Fire Office (EFO), Tunnel Ventilation Room, and structures like Pre-Engineered Building (PEB)
- PC8.** identify the common maintenance requirements for station building components such as false ceilings, false flooring, fire doors, and glasswork, explaining appropriate materials and methods for upkeep and repair to maintain safety and aesthetic standards
- PC9.** apply the correct procedures for maintenance work in public areas, including use of barricades to secure areas, adherence to the hot work permit system, and ensuring minimal disruption to passengers during repairs in operational stations
- PC10.** update and maintain station registers such as the Petty Repair Register, and categorize defects (A, B, C) as per Metro guidelines, conducting joint inspections with relevant wings and ensuring comprehensive documentation of maintenance activities

Duties and Responsibilities of Works Junior Engineer

To be competent, the user/individual on the job must be able to:

- PC11.** perform inspections for stations, colonies, depots, bridges, tunnels, and influence zones (viaducts & tunnels), noting observations in relevant registers and coordinating joint inspections with other departments per specified frequencies
- PC12.** record inspection findings in designated registers and document follow-up actions to ensure timely compliance
- PC13.** oversee routine and inspection-driven maintenance, prioritizing tasks that impact passenger safety, and ensure all works meet Metro standards

Viaducts/Bridge/Foundation/Steel Structure - Inspection & Maintenance

To be competent, the user/individual on the job must be able to:

- PC14.** inspect concrete and steel spans, documenting findings in the Bridge Inspection Register and following procedures for POT/PTFE and neoprene bearing checks per IRC 83 standards
- PC15.** examine bearings for shear and compression limits, referencing IRC codes, IRICEN, and RDSO guidelines for compliance
- PC16.** oversee viaduct maintenance activities, including structural repairs like jacketing, anti-carbonation painting, and drainage system repairs on elevated spans, according to the viaduct maintenance manual
- PC17.** record all observations and track compliance in maintenance registers to ensure follow-up on inspection and maintenance actions

Horticulture Works in O&M

To be competent, the user/individual on the job must be able to:

- PC18.** oversee horticulture maintenance for trees, shrubs, lawns, hedges, and seasonal flowers in Metro operations
- PC19.** follow the scheduled care and upkeep of horticulture items, ensuring seasonal maintenance of plants per Metro guidelines
- PC20.** apply Delhi Schedule of Rates (DSR) standards to horticulture work, maintaining quality and cost compliance

Qualification Pack

- PC21.** monitor horticulture contracts, ensuring that contractors meet Metro's requirements and schedules for landscape maintenance

Maintenance of Stations Elevated and Underground

To be competent, the user/individual on the job must be able to:

- PC22.** oversee the painting of steel structures, external and internal building surfaces, ensuring adherence to maintenance schedules and quality standards
- PC23.** inspect and maintain false ceiling and flooring systems, including repairs and replacements as needed to ensure functionality and aesthetics
- PC24.** oversee repairs and maintenance of flooring and cladding materials such as Kota stone, granite (dry and wet cladding), tiles, and yellow tactile surfaces to maintain durability and safety
- PC25.** oversee the replacement or repair of fire door accessories, including locks, panic bars, and related components, ensuring compliance with safety standards
- PC26.** maintain paving stones, kerb stones, and external façades of station buildings to preserve structural integrity and aesthetics
- PC27.** ensure the upkeep of public amenities, including regular maintenance of toilets and fittings in disabled-accessible toilets, addressing wear and tear promptly

Maintenance of Tunnels

To be competent, the user/individual on the job must be able to:

- PC28.** inspect and maintain tunnel cross-sectional elements, including walkways, fire pipelines, PTM lines, valves, cross passages, and doors to ensure functionality and safety
- PC29.** conduct detailed inspections of tunnels and underground station shafts (e.g., fire shafts, mid-tunnel shafts), documenting findings in the Tunnel Inspection Register
- PC30.** check and maintain cross-passage doors and other fire doors in tunnels, ensuring they are operational and compliant with safety regulations
- PC31.** identify and treat leakages in underground structures using waterproofing techniques, maintaining expansion joints to ensure structural integrity

Concrete Technology and Pre-stressed Concrete

To be competent, the user/individual on the job must be able to:

- PC32.** perform grading of aggregates and evaluate its importance, ensuring quality control of cement and water through standard testing methods
- PC33.** oversee accurate batching, mixing, placing, and consolidation of concrete, adhering to quality standards for durability and uniformity
- PC34.** implement appropriate curing methods (curing compounds, membrane curing) to ensure optimal concrete strength and longevity
- PC35.** apply suitable admixtures and manage the use of ready-mixed concrete to enhance workability and achieve project specifications
- PC36.** coordinate pre-stressing and post-tensioning, ensuring proper application and performance of pre-stressed concrete

Metro Building Specifications

To be competent, the user/individual on the job must be able to:

- PC37.** coordinate earthwork activities, ensuring adherence to Metro technical specifications for excavation, filling, compaction, and disposal of surplus earth

Qualification Pack

- PC38.** monitor concrete production and placement, ensuring compliance with Metro specifications for mix design, strength, and durability
- PC39.** inspect and oversee the installation and removal of formwork, ensuring it meets Metro standards for alignment, stability, and finish quality
- PC40.** verify reinforcement placement, including cutting, bending, and binding, as per Metro specifications for spacing, lap lengths, and anchorage
- PC41.** implement waterproofing measures for building structures, ensuring materials and methods conform to Metro standards for leak prevention and durability
- PC42.** inspect and ensure proper installation of roof sheeting materials, adhering to Metro guidelines for weatherproofing and load-bearing capacity
- PC43.** oversee road construction and maintenance, ensuring compliance with Metro specifications for subgrade, pavement layers, and surfacing
- PC44.** monitor building works, including masonry, plastering, and finishing, to ensure alignment with Metro technical specifications and quality standards

Water Supply

To be competent, the user/individual on the job must be able to:

- PC45.** inspect and manage water supply systems in elevated and underground stations, including bore wells, DJB supply, and PTM pipelines, ensuring functionality and compliance
- PC46.** oversee bore well redevelopment and maintenance of bore well pumps, coordinating with the E&M wing for seamless operation
- PC47.** conduct water sampling and testing per IS 10500:1991 standards, ensuring the quality of drinking water through periodic checks and external validation
- PC48.** coordinate cleaning of storage tanks and manage water distribution in stations, colonies, and depots, ensuring proper use of valves and meters in compliance with Metro's water policy
- PC49.** oversee the maintenance of water treatment plants, including softening plants and RO systems, as per contract BOQs, ensuring water quality in depots and staff quarters
- PC50.** inspect and maintain rainwater harvesting systems installed in depots, stations, and viaducts, ensuring optimal performance
- PC51.** coordinate maintenance of fire systems, including pipelines, specifications, fittings, and firefighting contract items, ensuring readiness and safety standards compliance

Sanitation

To be competent, the user/individual on the job must be able to:

- PC52.** inspect and maintain drainage and sewerage systems in stations, colonies, and depots, including surface drainage and sewer lines, ensuring effective operation and timely cleaning
- PC53.** oversee maintenance of sanitary installations such as septic tanks, soak pits, and bio-digesters, ensuring compliance with standard specifications for water supply fittings
- PC54.** coordinate the operation and maintenance of STP/ETP plants of various types and capacities installed in Metro, ensuring adherence to the maintenance contract BOQ and functionality standards

PTM Pipeline

To be competent, the user/individual on the job must be able to:

- PC55.** inspect the layout of PTM pipelines, including the configuration of the PTM pumping room and the various pumps installed

Qualification Pack

- PC56.** ensure compliance with specifications of pipelines used in Metro, including material type, pressure ratings, and durability
- PC57.** inspect various fittings and valves installed in the PTM pipeline system, ensuring their proper function
- PC58.** coordinate maintenance activities of PTM pipelines and related components as per typical maintenance contract BOQs, ensuring all operations meet performance and safety standards

Estimates

To be competent, the user/individual on the job must be able to:

- PC59.** prepare detailed estimates for Metro projects, incorporating DSR and NDSR items, ensuring alignment with Metro technical guidelines
- PC60.** conduct rate analysis, including a market survey of new items, and utilize methodologies like considering three latest LARs for accurate costing
- PC61.** submit estimates for vetting by the Finance wing, ensuring compliance with Metro's financial procedures and documentation standards

Tender and Contract for Works and Standard Specifications

To be competent, the user/individual on the job must be able to:

- PC62.** interpret and apply general and special conditions of contract in the preparation and execution of contract agreements
- PC63.** verify the proper submission of earnest money and security deposits as per tender guidelines
- PC64.** assist in tender evaluation by the tender committee, ensuring adherence to procedures for delayed or late tenders
- PC65.** monitor contractor classifications, manage advance payments, and ensure the application of liquidated damages in case of contract delays

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** construction activities for metro stations (underground and elevated), viaducts, and bridges, including girder types (U Girder, Segmental Box Girder, I-Girder, and special spans)
- KU2.** tunneling techniques: Cut & Cover, Tunnel Boring Machine (TBM), and NATM (New Austrian Tunneling Method)
- KU3.** station building components: SMR rooms, CCC, EFO, Tunnel Ventilation Room, and items requiring maintenance (false ceilings, flooring, fire doors, glasswork)
- KU4.** the procedures for maintenance activities in running stations: barricading, hot work permits, and maintenance registers like petty repair and defect categorization
- KU5.** station, colonies, depot, and bridge inspections: frequencies, recording observations in registers, and compliance
- KU6.** viaduct and special span inspections: knowledge of concrete and steel spans, bearing types (POT/PTFE, Neoprene), shear/compression limits, and viaduct maintenance manuals
- KU7.** the maintenance works - routine, inspection-based, and items affecting passenger safety, including drainage systems and anti-carbonation painting
- KU8.** tunnel inspection - walkways, pipelines, cross passages, fire doors, shafts, and addressing leakages in underground structures (waterproofing, expansion joints)

Qualification Pack

- KU9.** aggregate grading, cement testing, batching, mixing, curing, construction joints, and admixture usage
- KU10.** the pre-stressing, post-tensioning methods, and Metro technical specifications for earthwork, concrete, waterproofing, and building works
- KU11.** water supply systems in elevated and underground stations: bore wells, DJB supply, PTM pipelines, and water testing as per IS 10500:1991
- KU12.** the maintenance of water treatment plants (softening and RO systems) and rainwater harvesting systems
- KU13.** fire systems - pipelines, fittings, and maintenance activities, including firefighting contracts
- KU14.** surface drainage, sewer lines, sanitary arrangements, septic tanks, soak pits, and bio-digesters
- KU15.** maintenance of STP/ETP plants: types, capacities, and contract requirements
- KU16.** the layout of PTM pipelines, pumping rooms, and specifications of pipelines, fittings, and valves
- KU17.** maintenance contracts for water distribution and drainage systems
- KU18.** the preparation of estimates - DSR/NDSR items, rate analysis, market surveys, and LAR-based estimates vetted by the Finance wing
- KU19.** the principles of contract agreements: types, conditions, tender systems, and handling liquidated damages, advances, and contractor classifications
- KU20.** the horticulture schedules: trees, shrubs, lawns, seasonal flowers, and the Delhi Schedule of Rates (DSR) for horticulture contracts
- KU21.** the techniques for painting steel structures, buildings (internal and external), and maintaining flooring/cladding materials (Kota stone, granite, tiles)
- KU22.** the repair and replacement of fire doors and public amenities maintenance, including special fittings for disabled toilets

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** analyze challenges during construction, maintenance, and inspections and devise effective solutions to ensure seamless operations
- GS2.** convey technical information clearly, coordinate with teams, contractors, and stakeholders, and document observations and findings effectively
- GS3.** organize and prioritize tasks to meet deadlines for construction, inspections, and maintenance activities while ensuring quality standards
- GS4.** evaluate construction techniques, maintenance procedures, and materials to ensure compliance with Metro standards and optimize efficiency
- GS5.** work collaboratively with multiple teams, including engineering, operations, and contractors, to achieve project goals and maintain operations
- GS6.** adjust to new technologies, tools, and procedures related to construction, inspection, and maintenance activities
- GS7.** inspect and document structural components, maintenance records, and safety compliance to avoid errors

Logo
Not
Available



Qualification Pack

- GS8.** use appropriate software for design, documentation, inspection reports, and project planning to enhance efficiency and accuracy
- GS9.** follow and implement advancements in construction methods, materials, and safety protocols to improve job performance
- GS10.** build positive relationships with colleagues, contractors, and clients, ensuring effective communication and smooth execution of activities

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Overview of Construction Activities in Metro</i>	2	2	-	1
PC1. identify the main construction stages and components of underground and elevated metro stations, including excavation, foundation laying, and superstructure construction	-	-	-	-
PC2. identify the appropriate tools and equipment for constructing viaducts and bridges, including U girders, single and twin segmental box girders, and I-girders	-	-	-	-
PC3. identify special spans, including cantilever and extra dosed bridge spans, and the specific equipment and support structures required to construct and stabilize the spans	-	-	-	-
PC4. apply the cut-and-cover method and top-bottom and bottom-up techniques, understanding their applications in constructing metro tunnels within densely populated urban environments	-	-	-	-
PC5. identify the basic operations of Tunnel Boring Machines (TBMs), including the equipment setup, tunneling stages, and safety protocols essential for efficient and safe TBM usage in metro tunneling	-	-	-	-
PC6. apply the New Austrian Tunneling Method (NATM) in controlled ground excavation and use shotcrete for tunnel support, ensuring stability and safety in tunnel construction	-	-	-	-
<i>Metro Station Building - Elevated & Underground</i>	2	2	-	1
PC7. identify the functions and access points of key components of elevated and underground Metro stations, including system rooms like the Station Master Room (SMR), Central Control Center (CCC), Emergency Fire Office (EFO), Tunnel Ventilation Room, and structures like Pre-Engineered Building (PEB)	-	-	-	-
PC8. identify the common maintenance requirements for station building components such as false ceilings, false flooring, fire doors, and glasswork, explaining appropriate materials and methods for upkeep and repair to maintain safety and aesthetic standards	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. apply the correct procedures for maintenance work in public areas, including use of barricades to secure areas, adherence to the hot work permit system, and ensuring minimal disruption to passengers during repairs in operational stations	-	-	-	-
PC10. update and maintain station registers such as the Petty Repair Register, and categorize defects (A, B, C) as per Metro guidelines, conducting joint inspections with relevant wings and ensuring comprehensive documentation of maintenance activities	-	-	-	-
<i>Duties and Responsibilities of Works Junior Engineer</i>	1	1	-	1
PC11. perform inspections for stations, colonies, depots, bridges, tunnels, and influence zones (viaducts & tunnels), noting observations in relevant registers and coordinating joint inspections with other departments per specified frequencies	-	-	-	-
PC12. record inspection findings in designated registers and document follow-up actions to ensure timely compliance	-	-	-	-
PC13. oversee routine and inspection-driven maintenance, prioritizing tasks that impact passenger safety, and ensure all works meet Metro standards	-	-	-	-
<i>Viaducts/Bridge/Foundation/Steel Structure - Inspection & Maintenance</i>	2	2	-	1
PC14. inspect concrete and steel spans, documenting findings in the Bridge Inspection Register and following procedures for POT/PTFE and neoprene bearing checks per IRC 83 standards	-	-	-	-
PC15. examine bearings for shear and compression limits, referencing IRC codes, IRICEN, and RDSO guidelines for compliance	-	-	-	-
PC16. oversee viaduct maintenance activities, including structural repairs like jacketing, anti-carbonation painting, and drainage system repairs on elevated spans, according to the viaduct maintenance manual	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. record all observations and track compliance in maintenance registers to ensure follow-up on inspection and maintenance actions	-	-	-	-
<i>Horticulture Works in O&M</i>	2	2	-	1
PC18. oversee horticulture maintenance for trees, shrubs, lawns, hedges, and seasonal flowers in Metro operations	-	-	-	-
PC19. follow the scheduled care and upkeep of horticulture items, ensuring seasonal maintenance of plants per Metro guidelines	-	-	-	-
PC20. apply Delhi Schedule of Rates (DSR) standards to horticulture work, maintaining quality and cost compliance	-	-	-	-
PC21. monitor horticulture contracts, ensuring that contractors meet Metro's requirements and schedules for landscape maintenance	-	-	-	-
<i>Maintenance of Stations Elevated and Underground</i>	2	2	-	1
PC22. oversee the painting of steel structures, external and internal building surfaces, ensuring adherence to maintenance schedules and quality standards	-	-	-	-
PC23. inspect and maintain false ceiling and flooring systems, including repairs and replacements as needed to ensure functionality and aesthetics	-	-	-	-
PC24. oversee repairs and maintenance of flooring and cladding materials such as Kota stone, granite (dry and wet cladding), tiles, and yellow tactile surfaces to maintain durability and safety	-	-	-	-
PC25. oversee the replacement or repair of fire door accessories, including locks, panic bars, and related components, ensuring compliance with safety standards	-	-	-	-
PC26. maintain paving stones, kerb stones, and external façades of station buildings to preserve structural integrity and aesthetics	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure the upkeep of public amenities, including regular maintenance of toilets and fittings in disabled-accessible toilets, addressing wear and tear promptly	-	-	-	-
<i>Maintenance of Tunnels</i>	2	2	-	0.5
PC28. inspect and maintain tunnel cross-sectional elements, including walkways, fire pipelines, PTM lines, valves, cross passages, and doors to ensure functionality and safety	-	-	-	-
PC29. conduct detailed inspections of tunnels and underground station shafts (e.g., fire shafts, mid-tunnel shafts), documenting findings in the Tunnel Inspection Register	-	-	-	-
PC30. check and maintain cross-passage doors and other fire doors in tunnels, ensuring they are operational and compliant with safety regulations	-	-	-	-
PC31. identify and treat leakages in underground structures using waterproofing techniques, maintaining expansion joints to ensure structural integrity	-	-	-	-
<i>Concrete Technology and Pre-stressed Concrete</i>	1	1	-	0.5
PC32. perform grading of aggregates and evaluate its importance, ensuring quality control of cement and water through standard testing methods	-	-	-	-
PC33. oversee accurate batching, mixing, placing, and consolidation of concrete, adhering to quality standards for durability and uniformity	-	-	-	-
PC34. implement appropriate curing methods (curing compounds, membrane curing) to ensure optimal concrete strength and longevity	-	-	-	-
PC35. apply suitable admixtures and manage the use of ready-mixed concrete to enhance workability and achieve project specifications	-	-	-	-
PC36. coordinate pre-stressing and post-tensioning, ensuring proper application and performance of pre-stressed concrete	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Metro Building Specifications</i>	1	1	-	0.5
PC37. coordinate earthwork activities, ensuring adherence to Metro technical specifications for excavation, filling, compaction, and disposal of surplus earth	-	-	-	-
PC38. monitor concrete production and placement, ensuring compliance with Metro specifications for mix design, strength, and durability	-	-	-	-
PC39. inspect and oversee the installation and removal of formwork, ensuring it meets Metro standards for alignment, stability, and finish quality	-	-	-	-
PC40. verify reinforcement placement, including cutting, bending, and binding, as per Metro specifications for spacing, lap lengths, and anchorage	-	-	-	-
PC41. implement waterproofing measures for building structures, ensuring materials and methods conform to Metro standards for leak prevention and durability	-	-	-	-
PC42. inspect and ensure proper installation of roof sheeting materials, adhering to Metro guidelines for weatherproofing and load-bearing capacity	-	-	-	-
PC43. oversee road construction and maintenance, ensuring compliance with Metro specifications for subgrade, pavement layers, and surfacing	-	-	-	-
PC44. monitor building works, including masonry, plastering, and finishing, to ensure alignment with Metro technical specifications and quality standards	-	-	-	-
<i>Water Supply</i>	1	1	-	0.5
PC45. inspect and manage water supply systems in elevated and underground stations, including bore wells, DJB supply, and PTM pipelines, ensuring functionality and compliance	-	-	-	-
PC46. oversee bore well redevelopment and maintenance of bore well pumps, coordinating with the E&M wing for seamless operation	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC47. conduct water sampling and testing per IS 10500:1991 standards, ensuring the quality of drinking water through periodic checks and external validation	-	-	-	-
PC48. coordinate cleaning of storage tanks and manage water distribution in stations, colonies, and depots, ensuring proper use of valves and meters in compliance with Metro's water policy	-	-	-	-
PC49. oversee the maintenance of water treatment plants, including softening plants and RO systems, as per contract BOQs, ensuring water quality in depots and staff quarters	-	-	-	-
PC50. inspect and maintain rainwater harvesting systems installed in depots, stations, and viaducts, ensuring optimal performance	-	-	-	-
PC51. coordinate maintenance of fire systems, including pipelines, specifications, fittings, and firefighting contract items, ensuring readiness and safety standards compliance	-	-	-	-
<i>Sanitation</i>	1	1	-	0.5
PC52. inspect and maintain drainage and sewerage systems in stations, colonies, and depots, including surface drainage and sewer lines, ensuring effective operation and timely cleaning	-	-	-	-
PC53. oversee maintenance of sanitary installations such as septic tanks, soak pits, and bio-digesters, ensuring compliance with standard specifications for water supply fittings	-	-	-	-
PC54. coordinate the operation and maintenance of STP/ETP plants of various types and capacities installed in Metro, ensuring adherence to the maintenance contract BOQ and functionality standards	-	-	-	-
<i>PTM Pipeline</i>	1	1	-	0.5
PC55. inspect the layout of PTM pipelines, including the configuration of the PTM pumping room and the various pumps installed	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC56. ensure compliance with specifications of pipelines used in Metro, including material type, pressure ratings, and durability	-	-	-	-
PC57. inspect various fittings and valves installed in the PTM pipeline system, ensuring their proper function	-	-	-	-
PC58. coordinate maintenance activities of PTM pipelines and related components as per typical maintenance contract BOQs, ensuring all operations meet performance and safety standards	-	-	-	-
<i>Estimates</i>	1	1	-	0.5
PC59. prepare detailed estimates for Metro projects, incorporating DSR and NDSR items, ensuring alignment with Metro technical guidelines	-	-	-	-
PC60. conduct rate analysis, including a market survey of new items, and utilize methodologies like considering three latest LARs for accurate costing	-	-	-	-
PC61. submit estimates for vetting by the Finance wing, ensuring compliance with Metro's financial procedures and documentation standards	-	-	-	-
<i>Tender and Contract for Works and Standard Specifications</i>	1	1	-	0.5
PC62. interpret and apply general and special conditions of contract in the preparation and execution of contract agreements	-	-	-	-
PC63. verify the proper submission of earnest money and security deposits as per tender guidelines	-	-	-	-
PC64. assist in tender evaluation by the tender committee, ensuring adherence to procedures for delayed or late tenders	-	-	-	-
PC65. monitor contractor classifications, manage advance payments, and ensure the application of liquidated damages in case of contract delays	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0406
NOS Name	Coordinate construction, maintenance, and inspection in Metro projects
Sector	Logistics
Sub-Sector	
Occupation	Civil
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DMR/LSC/N0404: Conduct site visits for the inspection and maintenance of viaducts, stations, and treatment plants

Description

This unit covers the processes and standards for conducting site visits and inspections of Metro viaducts, bridges, station buildings, and water/sewage treatment plants (WTP/STP/ETP). It emphasizes defect identification, compliance with maintenance protocols, and coordination for corrective actions. The focus is on maintaining structural integrity, operational safety, and environmental standards through systematic inspection and documentation

Scope

The scope covers the following :

- Site Visit to Inspect Viaduct/Bridge
- Station Visit and Maintenance Overview
- Site Visit to WTP, STP, and ETP

Elements and Performance Criteria

Site Visit to Inspect Viaduct/Bridge

To be competent, the user/individual on the job must be able to:

- PC1.** ensure necessary tools, inspection forms, and safety gear are prepared and carried for the site visit
- PC2.** conduct detailed inspections of viaduct/bridge elements, including concrete and steel spans, bearings (POT/PTFE and neoprene), joints, and drainage systems, to identify defects or maintenance needs
- PC3.** record inspection findings accurately in the Bridge Inspection Register, noting observations such as shear/compression limits of bearings and structural condition
- PC4.** assess compliance of viaduct/bridge conditions with standards and best practices by referring to IRC, IRICEN, and metro manuals
- PC5.** recommend maintenance activities like repair of structural elements, drainage systems, and anti-carbonation treatments, and ensure proper follow-up through compliance recording

Station Visit and Maintenance Overview

To be competent, the user/individual on the job must be able to:

- PC6.** coordinate with JE/Works and other relevant staff to gather station inspection schedules, tools, and checklists
- PC7.** conduct detailed inspections of station components such as false ceilings, false floorings, external and internal painting, fire doors (including locks and panic bars), and other station building elements to assess their condition and maintenance needs
- PC8.** participate in joint inspections with officials from different wings, ensuring collaborative identification and prioritization of maintenance tasks
- PC9.** record inspection findings in relevant formats, including defects identified, category of maintenance needs (A, B, C), and proposed timelines for resolution, as per station inspection protocols

Qualification Pack

PC10. ensure recommendations from the joint inspection report are communicated and monitored for timely action and compliance with metro standards

Site Visit to WTP, STP, and ETP

To be competent, the user/individual on the job must be able to:

PC11. collect technical documentation, maintenance schedules, and operational logs for the Water Treatment Plant (WTP), Sewage Treatment Plant (STP), and Effluent Treatment Plant (ETP) to understand the scope of inspection

PC12. conduct inspections of key components, including pipelines, pumps, valves, treatment tanks, filtration systems, and chemical dosing systems, to assess functionality and compliance with standards

PC13. review the maintenance history, performance parameters, and upkeep of treatment processes, ensuring adherence to routine and preventive maintenance schedules

PC14. observe plant operations, including water purification, sewage processing, and effluent treatment procedures, to ensure compliance with environmental regulations and metro policies

PC15. document inspection findings, highlight any irregularities or maintenance needs, and propose corrective actions, ensuring alignment with plant operation protocols and maintenance contracts

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. viaduct and bridge structures, including concrete spans, steel spans, bearings (POT/PTFE, Neoprene), and drainage systems

KU2. inspection methods for structural integrity, identifying cracks, corrosion, or wear in components

KU3. the use of tools and equipment for viaduct/bridge inspection, including methods for bearing inspection (shear and compression limits)

KU4. the codes such as IRC 83 and manuals like IRICEN and RDSO publications for viaduct maintenance

KU5. the observations recording requirements concerning bridge inspection registers and preparing compliance reports for maintenance activities

KU6. station maintenance overview

KU7. station components requiring maintenance, such as:

KU8. the inspection station false ceilings for damage, loose fittings, and proper alignment

KU9. the inspection of station false floorings for identifying wear, cracks, and leveling issues

KU10. the types of fire doors, their lock systems, panic bars, and maintenance requirements

KU11. different types of internal and external paints, including anti-corrosion and fire-resistant paints

KU12. the PEB structures and glassworks in station buildings

KU13. the joint inspection protocols with officials from other wings, including frequency and mode of documenting findings

KU14. the reporting mechanisms, such as defect categorization (A, B, C), petty repair registers, and action plans

Qualification Pack

- KU15.** the collaborative inspection procedures
- KU16.** the roles of different departments in a joint inspection (e.g., electrical, civil, and mechanical wings)
- KU17.** the cross-functional systems like fire safety, ventilation, and public amenities
- KU18.** the methods to document observations in joint inspection reports, categorizing findings, and proposing corrective actions
- KU19.** the importance of ensuring compliance with maintenance protocols and following up on unresolved issues from previous inspections
- KU20.** WTP: Softening plants, RO systems, and processes for treating water to IS 10500:1991 standards
- KU21.** STP/ETP: Types, capacities, and treatment technologies like aerobic and anaerobic processes
- KU22.** the testing procedures for water quality and treated effluent
- KU23.** water treatment plant components requiring regular maintenance, such as pumps, valves, filters, and piping systems
- KU24.** the procedures for cleaning and redeveloping borewells, storage tanks, and distribution pipelines
- KU25.** the routine and breakdown maintenance activities in STP/ETP plants
- KU26.** the maintenance contract BOQs, schedules for inspections, and preparing reports based on findings during site visits

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** evaluate and analyze inspection data to identify potential defects or maintenance needs in infrastructure and plant systems
- GS2.** evaluate and analyze inspection data to identify potential defects or maintenance needs in infrastructure and plant systems
- GS3.** document and communicate inspection findings and recommendations clearly to relevant personnel to ensure timely follow-up actions
- GS4.** prioritize inspection tasks efficiently to ensure critical issues are addressed promptly within the inspection and maintenance schedules
- GS5.** identify and propose corrective actions for irregularities or faults discovered during inspections, ensuring compliance with standards
- GS6.** document inspection results accurately and recommended maintenance actions in appropriate formats for future reference
- GS7.** ensure that inspections and maintenance activities are in line with relevant standards, regulations, and organizational policies

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Site Visit to Inspect Viaduct/Bridge</i>	6	6	-	4
PC1. ensure necessary tools, inspection forms, and safety gear are prepared and carried for the site visit	-	-	-	-
PC2. conduct detailed inspections of viaduct/bridge elements, including concrete and steel spans, bearings (POT/PTFE and neoprene), joints, and drainage systems, to identify defects or maintenance needs	-	-	-	-
PC3. record inspection findings accurately in the Bridge Inspection Register, noting observations such as shear/compression limits of bearings and structural condition	-	-	-	-
PC4. assess compliance of viaduct/bridge conditions with standards and best practices by referring to IRC, IRICEN, and metro manuals	-	-	-	-
PC5. recommend maintenance activities like repair of structural elements, drainage systems, and anti-carbonation treatments, and ensure proper follow-up through compliance recording	-	-	-	-
<i>Station Visit and Maintenance Overview</i>	8	8	-	3
PC6. coordinate with JE/Works and other relevant staff to gather station inspection schedules, tools, and checklists	-	-	-	-
PC7. conduct detailed inspections of station components such as false ceilings, false floorings, external and internal painting, fire doors (including locks and panic bars), and other station building elements to assess their condition and maintenance needs	-	-	-	-
PC8. participate in joint inspections with officials from different wings, ensuring collaborative identification and prioritization of maintenance tasks	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. record inspection findings in relevant formats, including defects identified, category of maintenance needs (A, B, C), and proposed timelines for resolution, as per station inspection protocols	-	-	-	-
PC10. ensure recommendations from the joint inspection report are communicated and monitored for timely action and compliance with metro standards	-	-	-	-
<i>Site Visit to WTP, STP, and ETP</i>	6	6	-	3
PC11. collect technical documentation, maintenance schedules, and operational logs for the Water Treatment Plant (WTP), Sewage Treatment Plant (STP), and Effluent Treatment Plant (ETP) to understand the scope of inspection	-	-	-	-
PC12. conduct inspections of key components, including pipelines, pumps, valves, treatment tanks, filtration systems, and chemical dosing systems, to assess functionality and compliance with standards	-	-	-	-
PC13. review the maintenance history, performance parameters, and upkeep of treatment processes, ensuring adherence to routine and preventive maintenance schedules	-	-	-	-
PC14. observe plant operations, including water purification, sewage processing, and effluent treatment procedures, to ensure compliance with environmental regulations and metro policies	-	-	-	-
PC15. document inspection findings, highlight any irregularities or maintenance needs, and propose corrective actions, ensuring alignment with plant operation protocols and maintenance contracts	-	-	-	-
NOS Total	20	20	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DMR/LSC/N0404
NOS Name	Conduct site visits for the inspection and maintenance of viaducts, stations, and treatment plants
Sector	Logistics
Sub-Sector	
Occupation	Civil
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on a knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per the assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practicals for every student at each examination/training centre based on this criterion.
6. To pass the Qualification Pack, every trainee should score a minimum of 60% of aggregate marks to successfully clear the assessment.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DMR/LSC/N9901.Implement metro operations training and safety procedures	20	20	0	10	50	10
DMR/LSC/N0402.Coordinate the railway track maintenance and inspection	20	20	-	10	50	15
DMR/LSC/N0403.Coordinate the maintenance and inspection of LWR and associated track elements	20	20	-	10	50	15
DMR/LSC/N0405.Oversee accident and emergency response, track renewal, and implement inspection protocols	20	20	-	10	50	15
DMR/LSC/N0401.Use P.Way tools and equipment for track inspection maintenance	20	20	-	10	50	15
DMR/LSC/N0406.Coordinate construction, maintenance, and inspection in Metro projects	20	20	-	10	50	15
DMR/LSC/N0404.Conduct site visits for the inspection and maintenance of viaducts, stations, and treatment plants	20	20	-	10	50	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	160	170	0	70	400	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
QP	Qualification Pack
MC	Model Curriculum
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
MR	Metro Railway
SCM	Supervisor - Civil Maintenance
RR	Roles and Responsibilities
PWI	Permanent Way Inspection
TAP	Track Alignment Parameters
BCT	Ballasted and Concrete Tracks
CRC	Curve Radius and Cant
STL	Special Track Layouts
WED	Water Erosion and Drainage
BDS	Ballast Deep Screening
LWR/CWR	Long Welded Rail/Continuous Welded Rail
ERM	Emergency Response Management
SMP	Safety and Maintenance Protocols

Qualification Pack

HSL	Hand Signals and Lamps
STM	Small Track Machines
RRK	Rail Replacement Kit
ITK	Inspection Tool Kit
VIP	Viaducts and Infrastructure Projects
PCM	Project Cost Management
QCS	Quality Control Standards
WTP	Water Treatment Plant
STP	Sewage Treatment Plant
ETP	Effluent Treatment Plant
IPM	Inspection and Preventive Maintenance
IRF	Inspection Report Findings
CAF	Corrective Action Framework
DMRC	Delhi Metro Rail Corporation

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.